Final Design Review

Coyote Prosthetics - Multi-Part Foot Shell

ME 489

COYOTE
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Executive Summary

Coyote Prosthetics and Orthotics partnered with a senior engineering team at Boise State University to redesign a prosthetic footshell with the goal of improving patient independence, safety, and comfort while reducing long-term maintenance burdens for clinicians. Traditional footshells provide cosmetic coverage and basic traction but are difficult to install, prone to premature wear, and incompatible with sandal use or wet environments. These limitations restrict patient mobility and increase clinic workload. The redesigned multi-part footshell directly addresses these issues through a modular, additively manufactured system that prioritizes ease of installation, improved traction, water management, and compatibility with a wide range of prosthetic keels.

The final design consists of two TPU components, a forefoot and a heel piece, that interlock using a geometric seam, bolt-nut fasteners, and a T-rail system that prevents lateral shifting during gait. Internal TPU friction sleeves secure the shell to the keel without requiring adhesives or specialized tools, allowing patients and clinicians to install or replace the shell quickly. A sandal-compatible split toe expands footwear options, while integrated tread and drainage channels support barefoot use in wet environments such as showers, pools, and outdoor surfaces. The internal cavity and flexible sidewalls accommodate multiple keel geometries, ensuring compatibility across common prosthetic foot models used in the industry.

Prototyping demonstrated the feasibility of the design. FDM TPU prototypes validated the modular architecture, confirming that the shell could be installed without tools, maintained a secure fit during simulated gait, and provided adequate flexibility and comfort. A final Multi Jet Fusion (MJF) prototype was produced to evaluate manufacturability at production-grade resolution. While the MJF version confirmed that the design could be fabricated at high precision, it also revealed challenges including rail binding, increased stiffness, and higher-than-expected weight. These findings will guide future refinement and highlight the need for tolerance adjustments and weight-reduction strategies before large-scale production.

Cost analysis showed that single-unit MJF production currently exceeds the $100 target, but Coyote Prosthetics anticipates that bulk manufacturing through a nationwide partner will reduce per-unit cost to approximately $100. This positions the design to be economically viable at scale, especially when considering reduced warranty claims, simplified installation, and improved patient satisfaction.

Overall, this project represents a significant advancement in prosthetic footshell design. The multi-part architecture improves usability, enhances barefoot and sandal compatibility, and supports water-resistant performance, all while maintaining compatibility with existing prosthetic components. Although additional refinement is needed to optimize manufacturability and long-term durability, the design establishes a strong foundation for a next-generation footshell that aligns with Coyote Prosthetics' mission of improving quality of life for prosthetic users worldwide.

Problem Statement:

Coyote Prosthetics is a global distribution and manufacturing company for prosthetic limbs and prosthetic parts. Leg prosthetics are made of many different parts: a metal and fiberglass framework or keel that acts like the bone and ligaments in the body, and an outer, softer, polymer shell that goes around the keel which acts like the flesh of the foot. This foot shell is vital to the function of the limb as it allows the prosthetic to index with the world like a normal foot. Coyote Prosthetics manufactures and distributes prosthetic limbs globally. Their current foot shell design, a semi-rigid polymer encasing a keel, poses usability challenges. Technicians and users struggle to install and remove the shell for adjustments or replacements, and the shell wears quickly and slips during barefoot use. These issues reduce functionality and increase maintenance. A redesigned foot shell must be easier to install, compatible with varied keel shapes and sizes, and scalable for manufacturing, without over-constraining the design. Solutions must be easy to manufacture and scale to different sizes and shapes of prosthetic limbs and mounts. There is a lot of freedom given for the overall design, with the primary goal making it far easier to install a footshell than conventional methods.

Stakeholder Analysis:

- Prosthetic Limb End-User
    - Aspects of the solution that help
        - **Durability** - A longer product life will reduce the frequency of replacements, ensuring reliability, minimizing disruption from shell failure and decreasing overall lifetime costs.
        - **Safety** - A treaded design will increase traction on wet or smooth surfaces, reducing risks of slipping while barefoot.
        - **Accessibility** - A multi-part shell will provide increased accessibility for barefoot use, and for adjusting the foot while on-the-go.
        - **Comfort** - Improved material design and drainage will help keep the shell dry during use in wet environments (showering, swimming, or rainy conditions), ensuring shell fit stays consistent and comfortable to use.
    - Aspects of the solution that hinder
        - **Upfront Cost** - More durable shell materials could increase upfront costs even if lifetime costs decrease overall.
        - **Increased Weight** - A more durable design may add bulk or weight.
        - **Complexity** - Multi-part design increases the complexity and may require operational familiarization and training for proper use and adjustment.
        - **Biomechanical Impact** - Multi-part shell geometries could impact how forces are transferred through the foot, potentially influencing gait symmetry and introducing strain in the knee, hip or lower back due to compensatory movements.
- Company Owner (Coyote Prosethetics and Orthotics)
    - Aspects of the solution that help
        - **Client Experience** - Enhanced design featuring treads and accessibility features will improve client experience and accessibility and may offer a competitive edge in the prosthetics industry.
        - **Scalability** - Incorporating additive design and manufacturing can enable faster, large scale, and cost effective production while also making global distribution easier.

        - **Profitability** - A durable, efficient design will increase demand, help reduce complaints, returns or warranty claims, improving profitability.
    - Aspects of the solution that hinder
        - **Ease of Use** - If the product is more difficult to use and maintain due to complex part geometry, it may lead to reduced demand.
        - **Supply-Chain Risk** - Reliance on specific materials or manufacturing processes could create bottlenecks should supply issues arise, limiting responsiveness to demand.
        - **Regulatory Compliance -** Due to the medical nature of the product, regulation involvement is likely necessary. Failing to meet regulatory standards could hinder release to market.

- Coyote's Prosethic's International Clients/Distributors
    - Aspects of the solution that help
        - **Product Affordability** - Cost-effective and large scale manufacturing methods can increase opportunity for bulk purchases, and expand access to more end-user clients in their regions of sale.
        - **Product Reliability** - A durable, standardized product improves reliability by reducing variability between footshells and ensuring they are consistent with universally accepted shoe size standards. This helps streamline purchasing logistics and end-user fitting processes.
        - **Easy Replacement** - Additive manufacturing in combination with a multi-part design allows for easy replacement of worn or damaged portions of the footshell.
    - Aspects of the solution that hinder
        - **Climate Performance** - Additive manufacturing processes can have material limitations, which may cause performance issues under varying or extreme conditions (such as heat or UV exposure). This could impact performance across different local or international regions.
        - **Perception of Durability** - Hesitancy of purchasing may occur due to perception that additively manufactured products are cheap and unable to withstand long-term use.
        - **Support** - Increased design complexity may require more technical support which could be challenging to provide consistently across an international market.
- Clinician or Prosthetist
    - Aspects of the solution that help
        - **Compatibility** - A universal design will allow fit across different prosthetic components reducing compatibility issues and increasing availability and customization for patients.
        - **Choice Flexibility** - Design tailored for increased barefoot and aquatic use expands availability in footshell choice, providing clinician/prosthetists better options to care for their patient's individual needs.
    - Aspects of the solution that hinder
        - **Training Time** - Multi-part shell geometry may require additional time needed for training and patient demonstrations.
        - **Aesthetic Limitations** - Non-anatomical modelling could limit patient satisfaction.

- Manufacturers/Producers
    - Aspects of the solution that help
        - **Simplified Manufacturing** - Additive manufacturing allows for easier prototyping and rapid iteration for production.

        - **Market/Brand Recognition** - Being able to produce an innovative product can increase company prestige, help build client trust in the manufacturer's brand and increase demand for work.
    - Aspects of the solution that hinder
        - **Increased Setup Time -** Multi-part and tread features may require detailed setups or post-processing.
        - **Increased Quality Assurance** - Tighter tolerances to ensure proper fit between components may be required for a multi-part and multi-material design, increasing quality assurance requirements for production.

- Caregiver or Family Member
    - Aspects of the solution that help
        - **Easier Fit** - A footshell that is easier to put on while having access to adjust foot while wearing it will give patients more independence and reduce need for caregivers to assist.
        - **Modular Components** – Parts that can be replaced or adjusted without specialized tools reduce the need for clinic visits and save time for both patients and caregivers.
    - Aspects of the solution that hinder
        - **Time Commitment** – More parts or adjustments to properly maintain could increase daily caregiving tasks.
        - **Learning Curve** – If the device requires new skills or unfamiliar steps to use correctly, caregivers may need additional training, increasing early-stage workload and potential frustration.

Benchmarking:

BENCHMARK 1

Blatchford Navigator [1] https://www.blatchfordmobility.com/en-us/products/feet-ankles/navigator	Pros	Cons
	● Has split between first and second toe to allow wearing sandals ● Bears the appearance of feet which is a likely cosmetic desire from users ● Can be ordered in sizes ranging from a US womens 5 to a US womens 13	● Built in ankle joint makes it incompatible with other prosthesis ● Built in ankle joint also makes the shell more expensive to replace as the joint must be replaced too
The navigator is close to the industry standard, its primary drawback is the built in joint which is something that cannot be considered for our solution. Blatchford (the producer) does not indicate what the shell is made of, however, it is very similar to the shell currently used by Coyote Prosthetics, and likely has the same issue of a short lifespan, which is a current issue.		

BENCHMARK 2

Enviroshell Trustep [2] https://www.college-park.com/enviroshells-trustep	Pros	Cons
	● Many different variations depending on user needs including split between first and second toe ● Independently scaling foot width and length ● Multiple shell colors ● Tread is available for purchase to be added	● Shell is a foam blend leading to long term durability issues ● Shell must be completely removed to drain if it becomes waterlogged ● Lacks tread in base model
An improvement from the Navigator by removing the built in joint it makes the shell cheaper for the consumer and opens prosthesis modification for needs by the user's physician. In addition tread can be added which is similar to what is requested by the sponsor. However, it lacks a good way to handle waterlogging and still has the durability issues held by many other shells, meaning it still falls short of what the sponsor needs.		

BENCHMARK 3

<table>
<tr><td>Shelltread for Enviroshell [3]
https://www.college-park.com/shelltread-trustep</td><td rowspan="2">Pros

● Non-Slip
● Skid resistant
● Lightweight
● Water repellent (rubber)
● Softer rubber reduces scuffing</td><td rowspan="2">Cons

● Thin
● Shallow treads may easily wear down
● Cross-hatch design may trap small debris
● Hard to replace when worn down
● Specifically designed for Enviroshell footshell</td></tr>
<tr><td></td></tr>
<tr><td colspan="3">The shelltread for the Enviroshell is the current product offered to add anti-skid/slip to the Enviro-Shell Footshell. It is cross-hatched rubber that is applied by adhesive to the bottom of the footshell. It provides traction, is lightweight and water repellent, but because it is thin, it will wear down quickly and it is difficult to replace. Aspects of this solution could work for our design as it offers effective anti-skid and water repelling properties, but its limited durability and the difficulty of replacement would need to be addressed.</td></tr>
</table>

BENCHMARK 4

<table>
<tr><td>Ki Ecobe Modular Shoe [4]
https://www.core77.com/projects/66888/Modular-Self-assembled-Footwear-Designed-to-Put-the-Kibosh-on-Sweatshops</td><td rowspan="2">Pros

● Multi-component customization
● Easy replacement of individual pieces
● No adhesives (mechanically fastened)
● User-centered
● Lightweight
● Scalable
● Could be easily 3D printed</td><td rowspan="2">Cons

● Durability - (utilizes eco-friendly material which may lack strength)
● Complex (large number of components)
● No sandle-toe
● Waterlogging in wet environments</td></tr>
<tr><td> </td></tr>
<tr><td colspan="3">This modular shoe illustrates how effective a multi-part design can be for adaptability, and for customization for personal needs and aesthetics. A modular design enables adding or removing components as needed, but the eco-materials used may not be durable enough for all-day wear and tear. A high component count also makes it unnecessarily complex for a footshell design. Overall, adaptation of the upper shoe and a mechanically fastened sole with tread system, along with removal of unnecessary components could work well for our design.</td></tr>
</table>

BENCHMARK 5

Dorset Orthopaedic Lower Limb Cosmetic Silicone [5] https://www.dorset-ortho.com/offer/prosthetics/cosmetic-silicone/lower-limb-cosmetic-silicone-solutions	Pros	Cons
 	● Ultra-realistic design for client style-requests ● Easy-installation/snug fit using Velcro straps ● Silicone composition allows for waterproof benefits ● Parted-toe allows patients to wear sandals ● 3D-print compatible material	● Silicone composition lacks necessary tread ● Silicone composition will wear quickly at points of concern (heel and toe) ● Ultra-realistic design may bring up cost concerns for both production and clientele ● Velcro straps only offered for high-top design. Foot shell appears to lack same mechanism
The ultra-realistic design offers clientele an aesthetically pleasing option. The silicone also offers a cheap alternative to production. With that said, the silicone has been seen to wear in the heel and toe areas of previous foot shells. This wear fails to meet criteria concerning durability. In addition, the foot shell design appears to be one piece, meaning the installation would be just as difficult as current designs.		

BENCHMARK 6

Fillauer ProCover K3 - K4 Footshell [6], [7] https://fillauer.com/product/procover/ https://www.spsco.com/procover-footshell.html	Pros	Cons
 	● Polyurethane foam for comfort and durability ● Two-part shell allows for easy installation ● Waterproof up to 1 meter ● K3-K4 activity level (high level) ● 3D-print compatible material	● Lacks toe-separation for sandal-compatibility ● Not aesthetically pleasing ● Lacks treaded bottom
This simple design offers solutions to installation concerns above all. Additionally, the material claims to blend durability with comfort. However, the separation being along the sagittal plane of the foot brings up stability concerns. Without precise molding techniques, the prosthetic within the foot shell will be off balance, wearing the foam unevenly, accentuating those imbalances. Finally, the design fails to offer an aesthetic look. Overall, this securing technique could be a useful design for simplicity of production and installation.		

BENCHMARK 7

Kane - EVA Non-Slip Shoes [8] https://kanefootwear.com/blogs/kane-blog/are-eva-shoes-non-slip?srsltid=AfmBOoqx2SjxaUzikTJsFA5qYhSGYC8v72zOSzUGmmLowDWUA_-sVdEH 	Pros ● EVA foam for lightweight cushioning ● Shock absorption reduces joint stress ● Textured outsole improves slip resistance ● Flexible for natural foot motion	Cons ● Grip varies on wet surfaces ● Softness may reduce ground feel ● Compresses over time with wear ● Heat-sensitive under high temps ● Limited structural support ● May lack lateral stability
The EVA non-slip shoes provide a good model of durable material on the bottom with treads, and a softer more aesthetic material on the top. They demonstrate a lot of the qualities expected in our design such as slip resistance, durability, support, crack resistance- but there are also some flaws to acknowledge when adapting it to our design like compression over time, lack of stability, and limited support.		

BENCHMARK 8

Doc Marten - Buzz Sole [9][10] https://www.drmartens.com/eu/en_eu/guides/our-soles https://supportca.drmartens.com/hc/en-ca/articles/41106936273043-What-Are-Dr-Martens-Made-Of 	Pros ● Genuine leather for durability and structure ● Soft leather variants for flexibility and comfort ● Canvas and nylon for lightweight breathability ● PVC sole for oil and slip resistance ● Air-cushioned sole for shock absorption ● Distinct heel loop for easy pull-on function	Cons ● Stiff leather may reduce natural motion ● Patent finishes prone to cracking over time ● Canvas lacks water resistance and durability ● Heavy build in some styles increases fatigue ● Heel loop adds bulk without functional gain ● Buzz sole may prioritize aesthetics over support

The sole of Doc Martens offers effective slip resistance, making it a reliable choice in industrial settings such as food service and mechanical work. Constructed from PVC, a material readily available to the team through the EIS, this design provides durability and support. However, its weight may introduce inefficiencies when paired with the current (and interchangeable) internal metal foot prosthetic. Despite this limitation, the sole's performance characteristics can be strategically adapted to the lower shell of the footshell to meet the sponsor's requirements for slip resistance, structural integrity, and user support.

BENCHMARK 9

College Park: Terrain K2- K4 [11] https://www.college-park.com/terrain 	Pros ● Closely mimics foot dynamics in a wide range of situations (Running, Hiking, Walking) ● Standard for the industry ● Scalable to different foot sizes	Cons ● Limited mounting options ● Rough on footshells ● Good in all conditions, but not specialized for specific actions. ● Fiber glass spring boards splinter over time
While not specifically this brand and model, this type of prosthetic is standard for prosthetics for every level of mobility. This type of prosthetic is what we will be initially modeling our footshell to attach to. A good balance of shock absorption and spring		

BENCHMARK 11

Ottoblock - Sprinter Blade Pads [12] https://www.ottobock.com/en-us/product/1E90-133566 	Pros ● Example of a treaded surface designed for prosthetic use ● Advanced linking system keeps pad attached without screws or straps, just friction ● Materials built to withstand harsh running conditions	Cons ● Built to attach to a running blade ● Material is meant to withstand track running and outside use, possibly not safe for carpet and wood flooring. ● Stress distribution in sprinting is different for regular wear, and may not provide enough grip or cushion if similar materials are used.

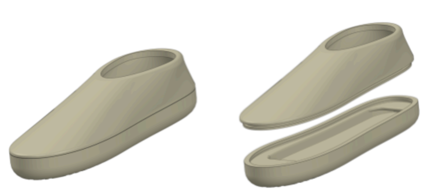
The Ottoblock blade pads are a different example of tread used for a prosthetic foot, one that is attached in a way that does not require screws being drilled into the prosthetic itself. Though very different from what we intend to make, the material used and techniques for attachment may be applicable to a footshell.

BENCHMARK 12

<table>
<tr><td>Pro-Flex Tera Water-Proof Footcover [13]
https://media.ossur.com/ossur-dam/image/upload/product-documents-global/Pro-Flex-Terra_Catalog_en-us.pdf

Pro-Flex Terra – Sandal Fix Installation (Clinician Fabrication Tutorial): [14]
https://www.youtube.com/watch?v=ajOfVe0IflM</td><td rowspan="2">Pros
Hinged back allows easy insertion of keel
Can access ankle easily
Waterproof
Cosmetically appealing</td><td rowspan="2">Cons
No sandal toe/wearing sandals requires drilling
No traction on bottom of shell</td></tr>
<tr><td></td></tr>
<tr><td colspan="3">This is a popular, high-quality foot-shell that is waterproof and has a relatively simple back-opening design to allow easy insertion of the prosthetic keel. This model, however, lacks an easy way to wear open toe footwear, with a clinician tutorial showing permanent drilling being required to fix the shell to the sandal. Waterproof material and a hinge or button style back to allow keel insertion could be viable for our multi-part footshell design.</td></tr>
</table>

BENCHMARK 13

<table>
<tr><td>Prosthetic Footshell - Boise State University Senior Design 2023 [12]
https://drive.google.com/file/d/1H6_o9Lqf6D2tja2i9IZPJKjRCqN8EsRE/view?usp=sharing.</td><td rowspan="2">Pros
Simple for additively manufactured design
Same relevant specifications as current design</td><td rowspan="2">Cons
Minimal structural support
Shell is a foam leading to long term durability issues
Lower shell must be removed to drain
Lacks tread in base model
Connection points susceptible to shear force.</td></tr>
<tr><td></td></tr>
<tr><td colspan="3">The Boise State Senior design footshell project from 2023 is simple, which is beneficial for additive manufacturing and compatibility with standard prosthetic specifications. However, it suffers from minimal internal support, a foam-blend construction prone to durability and water-retention issues, lack of integrated tread, and connection points vulnerable to shear. Generalizing the overall shape and adding tread and a toe could be beneficial as a starting point for a multi-part design which will be additively manufactured.</td></tr>
</table>

Needs:

Need ID	Description of Need	Justification	Required
1	Design installation must be safe for patient and practitioner	Requested by Client	Y
2	Design must support patient weight	Requested by Client	Y
3	Design must allow the user’s gait	Requested by Client	Y
4	Design must be multiple pieces	Requested by Client	Y
5	The design must be durable	Requested by Client	Y
6	Design must have treaded bottom	Requested by Client	Y
7	Design must be simple to install	Requested by Client	Y
8	Footshell design must be functional with or without a shoe	Requested by client	Y
9	Design must securely fit the keel	Requested by Client	Y
10	Design must be compatible with sandals	Requested by Client	Y
11	Design must be compatible with multiple prosthetic models	Requested by Client	Y
12	Design must utilize additive manufacturing	Requested by Client	Y
13	Design must be able to handle water exposure	Requested by Client	Y
14	Design process must be within allotted budget	Requested by Client	Y
15	Design must not make alterations to the prosthetic	Requested by Client	Y
16	Design must be aesthetic	Requested by Client	N

Background Information and Research:

Company Overview

Coyote Prosthetics is a research and development company that strives to enhance the lives of individuals through innovative orthopedic and prosthetic (O&P) solutions. Their mission is to design products that deliver the comfort, functionality, and quality of care they would want for their own friends and family. Coyote’s products are trusted by O&P practitioners across the United States and serve a wide range of patients, including amputees, individuals born with physical disabilities, and those with other medical needs [15]. The company offers a diverse portfolio of product lines tailored to meet the varying demands of the O&P community. One of Coyote's innovations is its Vacuum Technology line, featuring products such as the *Smartpuck, Airpuck,* and *Zeropuck* [15]. This technology implements continuous vacuum (tissue perfusion) to enhance blood circulation and promote residual limb health. It is the only prosthetic suspension system shown to physiologically improve limb condition by drawing blood, lymph, and nutrients into the residual limb, which results in stronger, healthier tissue [15].

Another product family designed by Coyote Prosthetics is the prosthetic locking systems, which include the *Air-lock*, *Drop-In Lock*, *Easy-Off Lock*, *Suction Lanyards*, *Locking Lanyards*, and *Lanyard Puck* [15]. These components use a lightweight dual-suspension pin system that merges the comfort of suction suspension with the security of pin suspension. The result is an airtight seal that minimizes air intrusion at the socket base, reducing pistoning and improving overall fit and stability [16]. While Coyote Prosthetics specializes in locking components and vacuum suspension systems, it operates in a competitive field alongside companies such as Össur, Naked Prosthetics, Ottobock, Blatchford, Hanger Inc., and Ohio Willow Wood. Among these, Össur and Ottobock stand out as global leaders, particularly in the premium segment of prosthetics and orthopedics. Despite not being a top-tier multinational company, Coyote distinguishes itself by offering durable, cost-effective, and forward-thinking prosthetic solutions [15]. Their products are especially appealing to clinics and practitioners seeking reliable alternatives to the major brands, without compromising on innovation or performance.

Problem Definition: Limitations of Current Foot Shells

One of the core challenges in redesigning the prosthetic footshell lies in addressing the limitations of current models while balancing usability, durability, and compatibility. Traditional footshells placed over a prosthetic foot keel wear out rapidly, especially at high-stress zones like the ball of the foot and heel, often failing within six months of use. These areas endure the most force during walking, making material selection critical. The existing design also poses usability concerns: it's difficult to install without a shoehorn and has even caused user injury during application. To improve functionality, the new design must incorporate treaded surfaces for barefoot use in wet environments, such as showers or pools, and feature a split toe to accommodate sandals. While the shell can consist of more than two parts, simplicity remains key for ease of use. A multi-material approach is promising, with a tougher bottom for grip and wear resistance and a softer upper for comfort and aesthetics. The design does not need to mimic anatomical features and may lean toward a futuristic look if it enhances appeal. It must also fit a size 11-12 US men shoes and work for both left and right feet, with the left foot used for testing. Compatibility with various prosthetic feet, including those from other manufacturers, requires added side clearance and adjustable set screw access. Since the metal prosthetic foot cannot be modified, the shell must adapt around it. With access to a 3D scanner but no in-house printing, the team will outsource production, starting with a single test print and scaling to 100 units as needed.

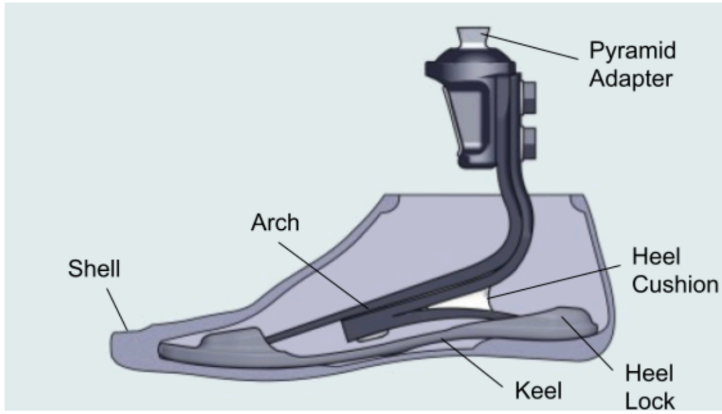


Figure 1. *Foot prosthetic keel showing the geometries placed inside a typical footshell.*

Review of 2023 Senior Design Efforts

In 2023, a senior design team at Boise State University collaborated with Coyote Prosthetics to develop a new foot shell prototype aimed at improving durability, traction, and ease of use. Their concept featured a two-piece design: a rugged bottom shell for barefoot functionality and a soft, aesthetic top cover [16]. The team selected polyurethane for its high-friction properties and incorporated a grommet-based attachment system to simplify installation and removal. Their design also addressed water exposure, split-toe compatibility for sandals, and cost constraints, keeping the prototype under a $520 budget.

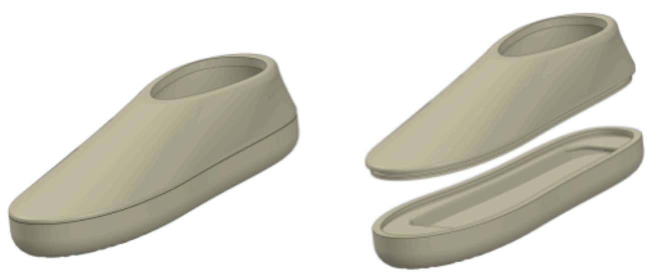


Figure 2. *Isometric view of the foot shell designed by 2023 Boise State Senior Design Group [16].*

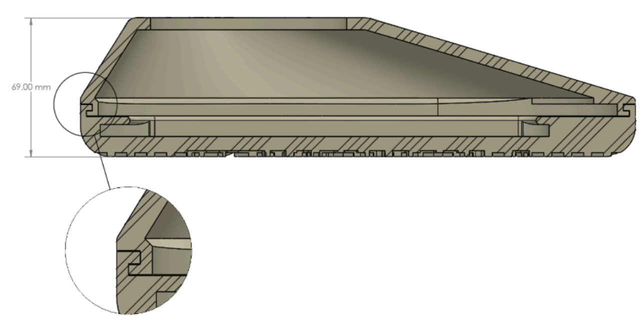


Figure 3. *Cross-section view of the foot shell designed by 2023 Boise State Senior Design Group [16].*

Despite these innovations, the project faced several limitations. The shell showed signs of premature wear during walking trials, particularly in high-stress zones like the toe-off and heel areas [16]. Installation remained difficult for users, with some requiring tools or assistance to avoid injury. Fitment issues emerged over time, as the shell loosened and failed to maintain a secure connection with the prosthetic foot. Additionally, noise generated between the shell and the metal keel detracted from user comfort. These challenges highlighted the need for improved material layering, better fatigue resistance, and more precise internal geometry. These insights now inform the next phase of design.

User Needs and Functional Requirements

The redesigned prosthetic footshell must meet a wide range of functional demands to support daily life for lower-limb amputees. Users rely on the footshell not only for cosmetic coverage but also for protection, traction, and comfort across varied environments [18]. It must perform reliably both barefoot and with shoes, including in wet conditions such as showers, pools, and outdoor terrain. This requires treaded surfaces to prevent slipping and materials that resist water absorption, microbial growth, and degradation from UV exposure. Ergonomically, the shell must accommodate natural gait mechanics and distribute pressure evenly, especially in high-impact zones like the heel and forefoot [18]. Biomechanically, it should allow for smooth rollover during walking and provide enough flexibility to avoid impeding motion. To do this the shell should fit the keel well and be made of a material that allows some flexibility. The inclusion of a split sandal toe expands usability, enabling wear with open-toed footwear. Additionally, the shell must be compatible with different potential prosthetic foot models and shoe sizes, with enough lateral clearance to fit securely without excessive compression. Use-case scenarios include walking on uneven ground, transitioning between wet and dry surfaces, and wearing only the treaded bottom when barefoot. These requirements demand a balance of durability, adaptability, and user comfort, which ensures the footshell enhances mobility rather than limiting it.

Materials and Additive Manufacturing Technologies

Additive manufacturing, commonly known as 3D printing, is a process of creating objects by building them layer by layer from digital models, typically using materials such as polymers, metals, or ceramics. This project relies on additive manufacturing to produce a multi-part prosthetic footshell that balances durability, comfort, and cosmetic appeal. During the research phase, our senior design team will utilize Fused Deposition Modeling (FDM) 3D printing due to its speed and accessibility [19]. However, final production may require more advanced methods such as Selective Laser Sintering (SLS) or Multi Jet Fusion (MJF), which offer superior mechanical properties, surface finish, and multi-material capabilities [20, 21]. These methods, while more effective, face accessibility limitations within the Engineering Innovation Studio (EIS). MJF and SLS are newer industrial-grade 3D printing technologies commonly used for final prototype designs [20]. They provide high productivity and material reusability. [20] Their mechanical strength and dimensional accuracy significantly surpass those of FDM, making FDM a suitable rapid solution for evaluating sizing and practicality, but not ideal for final production [19]. Material selection is critical: the bottom portion must withstand high-impact forces at the heel and ball of the foot, resist wear over time, and provide reliable traction in wet environments, while the top shell should be softer for comfort. A multi-material approach is ideal, using a treaded polymer for the sole and a flexible, aesthetic material for the upper shell.

An additional aspect to consider when selecting a material for 3D printing is shore hardness. Shore hardness is an important factor because it indicates how soft or rigid the final part will be. Materials with lower Shore hardness values can bend, compress, or absorb impact. Higher Shore hardness materials, like rigid PLA or ABS, offer greater stiffness and dimensional stability, which is beneficial for structural parts and mechanical assemblies. Understanding the required level of flexibility or rigidity helps ensure the chosen material will perform as intended. A general chart comparing shore hardness levels is shown in Figure 4.

Figure 4. *Comparison of Different Object Shore Hardness Levels [52]*

A potential material that could be applied for the shell are elastomers, which are a natural or synthetic polymer commonly used in biomedical designs to replicate natural skin tissue [22]. Polymers have long been successful in biomedical applications due to their adaptable physical and chemical properties, but elastomers are particularly relevant here[22]. Elastomers like silicone and polyurethane (PU) are ideal candidates because they combine flexibility, stretchability, and biocompatibility, closely mimicking the properties of human skin [22]. Both materials are recognized by ISO 10993 and the FDA for safe contact with skin, having passed cytotoxicity, irritation, and sensitization tests [22], [23]. Silicone, a chemically crosslinked thermoset elastomer, and PU, a physically crosslinked thermoplastic elastomer, offer distinct advantages: silicone provides long-term biostability, while PU offers mechanical strength and reprocessability [22]. Their hyper-elastic behavior allows for high strain recovery, making them suitable for wearable prosthetics that must endure repeated loading and deformation.

Out of common PU's, Thermoplastic Polyurethane (TPU) would be a prime material candidate for additively manufacturing the foot-shell components through MJF or SLS processes. TPUs have optimal flexibility and shock absorption due to their high ductility while being able to produce intricate geometries and fine detail [24]. Build orientation will be an important consideration throughout the design, as printing on-edge can enhance deformability and increase overall flexibility and strength. TPU's mechanical properties provide a strong foundation for durable, customized prosthetic footwear while also supporting ergonomic and aesthetic goals, which pushes it forward as a frontrunner for a material choice applicable to this design. Powders such as the Ultrastint TPU01 are widely available with uses in sport and leisure products as well as footwear [25]. In a recent study, TPU printed foot orthoses withstood approximately 700,000 loading cycles without visible surface damage or permanent deformation, and maintained their structural integrity after more than 1,400,000 steps of daily walking [26]. This level of durability is particularly encouraging for a footwear application, and could be utilized for both the bottom or sole of the footshell, and/or the top portion of the shell which is subject to less stress overall than the sole. Material properties for TPU are shown in Table 1 below, with a highlight being an impressive ultimate tensile strength of 8.6Mpa.

Property	Value, metric	Value, Imperial	Type
Density	1.1 g/cm3	1.1 g/cm3	-
Elongation at Break	220%	220%	-
Hardness	88	88	Shore A
Ultimate Tensile Strength	8.6 MPa	1.2 ksi	-

Table 1. *Xometry TPU Material Properties [53].*

A final option is to use Nylon as the material. Nylon is much more rigid than the silicone and PUs, providing less flexibility but instead offering high structural strength, quick and precise results and minimal material waste [27]. It is an alternative to injection molding and can be printed utilizing both MJF and SLS manufacturing techniques [28]. However, a key downside would be water absorption, especially in combination with SLS printing. Nylon absorbs and retains water. In addition, a frequently used material in SLS printing is a polyamide, which also absorbs moisture [29]. This has the potential to affect mechanical strength and produce swelling. Further consideration can include specific Nylon polyamides. Nylon PA11 is used for high ductility and impact resistance, however, it only comes in grey but can be dyed darker [27]. PA12 is an SLS based material and offers extreme resilience to friction. It is printed white, but can be dyed any color [28]. Overall, PA11 and PA12 offer humidity absorption of roughly 0.7-0.8% which is much less than other polyamides [27]. This offers reassurance as a contender for material consideration. Overall Nylon could be used practically to create a rigid, impact resistant frame that could be paired with softer materials for aesthetics and replaceable tread, as long as absorption is considered.

Potential Manufacturing Partners for Additive Prototyping and Production

Several companies offer advanced manufacturing abilities that align with the materials and processes used in prosthetic component design. For flexible thermoplastic polyurethane (TPU) parts, Xometry provides Multi Jet Fusion (MJF) services using Ultrastint TPU01 and other elastomers [30]. Their platform supports rapid prototyping and short-run production with fast turnaround times and nationwide reach [30]. For structural nylon components, Jawstec offers both Selective Laser Sintering (SLS) and MJF printing, specializing in high-strength, precision parts suitable for load-bearing applications [27],[28]. Based in Idaho, Jawstec also provides CAD support and post-processing services [31]. For silicone-based components, Sherpa Design, an official Carbon partner located in Portland, Oregon, offers Digital Light Synthesis™ (DLS) printing using biocompatible elastomers like SIL 30 [32]. Their services include design optimization and surface texturing, making them ideal for producing soft, durable prosthetic interfaces [32]. These vendors collectively support a range of material needs from flexible inserts to structural shells and soft liners, making them strong candidates for prototyping and low-volume production [27–32].

Material Selection Based on Anatomical Density

To enhance realism and biomechanical performance, the upper portion of the prosthetic foot shell should reflect the mass and density characteristics of a biological foot. According to anatomical research data, the human foot comprises approximately 1.45% of total body mass and has an average mass density of 1.10 g/cm³ [33]. This density is consistent with other lower limb segments such as the calf (1.09 g/cm³) and thigh (1.05 g/cm³), indicating that soft tissue in the leg is relatively uniform in its mechanical properties [33].

For the upper shell, selecting a material with a similar density ensures that the prosthetic foot behaves naturally during gait cycles, especially in terms of inertia and balance. Thermoplastic elastomers (TPEs), flexible polyurethanes, or silicone-based compounds can be tuned to match this density range while offering the softness and compliance needed for aesthetic realism [34]. This alignment minimizes the perceptual and physical disconnect between the prosthetic and the user's body, particularly when barefoot or wearing open footwear.

In addition to density matching, TPEs offer a range of mechanical properties that support functional integration [34]. Depending on the blend composition and processing method, TPEs can achieve Shore hardness values from 3A to 85D, tensile strengths suitable for load-bearing applications, and elongation capabilities that mimic soft tissue flexibility [34]. Their fatigue durability and low compression set make them ideal for repeated stress cycles, such as those experienced during walking or running [34]. Furthermore, TPEs exhibit good tear resistance and abrasion performance, which helps maintain shell integrity over time [34].

Finally, matching anatomical density supports better integration with the lower shell and keel, reducing stress concentrations and improving overall fatigue life. By basing material selection on biological data, the design achieves a more lifelike feel, improved proprioception, and enhanced user comfort, without compromising durability or manufacturability.

Design Integration and Compatibility

Coyote Prosthetics specializes in modular, high-performance prosthetic components,this includes their design of three-prong connectors and zero-clearance locking systems like the *Summit Lock* and *Air-Lock* [17] [35] and are shown in Figures 5 and 6 below. These systems are designed for secure suspension and rotational control, meaning the outer foot shell must maintain precise tolerances to avoid interfering with these functions [17] [35]. The shell should allow access to set screws and accommodate the geometry of Coyote's metal keel, which will be provided by Coyote Prosethetics for further design analysis.

Figure 5. *Coyote Prosthetics & Orthodontics Summit Lock mechanism [17].*

Figure 6. *Coyote Prosthetics & Orthodontics Air-Lock mechanism [17].*

To ensure compatibility across current and future models, the shell should follow ISO 10328 standards for structural integrity and fatigue resistance [36]. A slightly oversized internal cavity with flexible sidewalls or compression zones can support multiple keel designs while maintaining a snug, secure interface. Multi-material layering, rigid in high-wear zones, like the heel and toe-off, and flexible elsewhere, can help absorb forces without negatively affecting fit. The design must also support water drainage and barefoot traction, aligning with Coyote's emphasis on real-world usability and active lifestyles. By aligning with their modular philosophy and mechanical standards, the new shell can become a durable, adaptable component in Coyote's prosthetic ecosystem.

Installation and Modularity Considerations

Installation and modularity considerations for user-friendly prosthetic design must prioritize ease of installation, secure fit, and long-term adaptability. Feedback from users and Coyote Prosthetics highlight persistent challenges with current footshell models, particularly in high-stress regions like the ball of the foot and heel, areas that endure the greatest forces during walking. These components tend to wear out quickly, compromising both comfort and performance. Additionally, the installation process often requires excessive force or external tools like shoe horns, which has led to user injuries and frustration. Best practices suggest integrating tool-free attachment mechanisms, such as snap-fit interfaces, that allow for safe, intuitive assembly and disassembly. Drawing from recent research on the individual daily users of prosthetics, especially those intended for in-home use, can ensure that prosthetic components are not only durable and replaceable but also tailored to real-world conditions and user capabilities.

Cosmetic Appeal and Comfort Factors

Cosmetic appeal and comfort must be deeply intertwined in prosthetic design, especially when it comes to user satisfaction and long-term wearability. A softer upper shell plays a strong role, not just in improving comfort by reducing friction and pressure points, but also in enhancing the aesthetic quality of the device. This matters more than it might seem: recent research has shown that prosthetic appearance can significantly influence emotional connection, social confidence, and daily wear compliance of individuals who use prosthetics [37]. A 2023 mixed-methods study published in *Technology and Innovation* emphasizes that users often prioritize cosmetic realism and personalization alongside functional performance. When a prosthesis reflects a user's identity and lifestyle, it transitions from a clinical tool to a personal artifact. Personalization trends, such as custom colors, textures, and stylistic choices,allow users to express ownership and individuality, reinforcing psychological comfort and social integration [37]. This design philosophy also reflects Coyote Prosthetics' broader goal: to create prosthetic devices that foster realism and emotional connection between the user and their limb [15]. Their commitment to personalization and aesthetic integration goes beyond surface-level appearance, it's about helping users feel seen, supported, and confident in their daily lives [15]. By aligning cosmetic features with individual preferences, Coyote reinforces the idea that prosthetics should not only restore function, but also restore identity.

Furthermore, a cosmetic recommendation we received directly from Coyote Prosthetics was the addition of a toe slit, tailored specifically for a customer (and employee) who frequently wears sandals. This adjustment enhances the visual realism of the prosthetic, helping it blend more naturally with everyday footwear and reinforcing a sense of normalcy. It's a small but meaningful change that improves user satisfaction by aligning the device with personal style and lifestyle needs.

Ultimately, cosmetic design is not a superficial concern, it's a functional priority that supports emotional well-being, social participation, and long-term device acceptance. As the study concludes, appearance-related factors are central to how users perceive, adopt, and emotionally connect with their prosthesis [37].

Budget Constraints and Resource Planning

To maximize the amount of production and iterations with the limited budget, initial prototyping will likely need to rely on FDM printing due to its speed, affordability, and accessibility within the EIS, making it ideal for early iterations focused on geometry, fit, and basic usability [19]. This approach allows for rapid feedback and refinement without exceeding the $2,000 project budget. As the design process matures, higher-end additive manufacturing methods like SLS and MJF should be considered for final passes, especially in load-bearing regions such as the ball of the foot and heel, where mechanical stress is highest during walking. These advanced methods offer improved strength, surface finish, and durability, supporting long-term performance goals [20], [21]. The budget will also influence material selection, initially favoring cost-effective options like PLA or PETG for early-stage testing, while reserving premium materials for final components. This tiered strategy will balance affordability with functionality, allowing for iterative development that reflects real-world user's needs and constraints.

Problem Analysis:

In order for this project to be considered a success there are a number of checks that need to be passed, all at different levels of the design and function. Most importantly, our design must withstand comparison to existing prosthetic models by surviving a K4 activity level classification (patients engaging in "high impact activities or sports" [22]) and a 3-million cycle durability test. Our patient weighs 250 pounds, 50 pounds higher than the average human male. This is all background to the specific reason we are working on this project, which is to make the installation of the footshell far easier. Ease of use is not really something that can be quantified, but seeing as current shells require a tool for installation, our design goal is to eliminate specialized tools by creating a multipart shell that can be attached without external equipment.

The final set of bars we want to clear is prototype testing. The ISO 10328 standard, which is the internationally recognized durability test for lower-limb prosthetics, is both expensive and time-consuming. To manage resources effectively, we will first conduct a 10,000-step screening test to identify early wear. This preliminary test will not replace the 3-million cycle requirement but will serve as an indicator of whether the design merits full qualification. This approach helps avoid wasting time testing a device that shows failure under basic daily activity. As the project requires additive manufacturing, many of the potential materials lack established fatigue strengths, making direct fatigue calculations impractical and requiring assumptions to be made.

Footshell "Standards"

There are no industry or company standards for the performance of footshells, so for the final stage of testing to determine the success of the design is to survive a ISO-10328 prosthetic test. This is an international standard test for the entire lower-limb prosthesis, not just the footshell, and it uses combined loading to simulate 3 million steps. This is the final test we could hope to pass that would qualify the design as a viable product for international shipping. Additionally we'll need to make sure the design is not made of materials that are outlawed in the European Union.

New Design Requirements

For this project to be a success in our Senior Project we need to achieve the initial goal of this project: Making a multipart footshell that is easier than existing shells to attach to a prosthetic keel. If this primary usability goal is not met, the project will be considered unsuccessful regardless of other achievements.

The ultimate goal is to pass the ISO 10328 test, but this is an expensive and time consuming process. For the sake of prototype testing, a preliminary 10,000-cycle test will be conducted as a screening step, with the understanding that this is not equivalent to the full 3-million cycle requirement. The specifics of the test environment cannot yet be determined at this stage in the project, but it will need to survive 250-500 pounds of vertical force (varying depending on conditions like walking/running) and 12-60 pounds of shear force [39]. In a test rig that is cycling a vertical force, the ground that the footshell makes contact with can be angled accordingly to simulate the shear force. For heel testing, a contact angle of 15 degrees, while the forefoot will be tested on a surface angled at 20 degrees. These angles are chosen to approximate heel strike and toe-off conditions during gait, rather than static standing loads. This should be a good simulation of a combined loading test that can act as a substitute for the full ISO 10328 standard.

Additional Test Requirements

The footshell is required to function in most ways like a replacement for a natural foot. This calls for the footshell to have a tread and not pose a slipping risk in wet conditions like a shower. For the requirement of adding tread to this design, it has been kept relatively simple as the footshell does not need to act as a shoe or have detachable tread. A simple design inspired by nonslip tread will be used. Additionally, the footshell also needs to be compatible with sandals, specifically flip flops, and have a drain hole in order to clear itself of water without having to take off the shell. Similar to existing footshells, the new design needs to be compatible for a plethora of different keel designs. Slip-resistance testing will specify the type of surface used such as linoleum, concrete, or asphalt to avoid ambiguity in performance evaluation.

Equations

Given the testing configurations used to verify the repeated loading and slip-resistance parameters of the design, preliminary calculations were conducted in accordance with these specific testing rigs. The first parameter considered was the prosthetic foot provided by COYOTE Prosthetics. The contact surface area of the prosthetic foot with the ground was measured to be 2.6575 in^2 prior to loading. This measurement along with the applied load were used to derive the average contact stress. In accordance with ISO 10328, test loading level P6 outlines an applied axial force of 4880 N = 1096.629 lbf that would be experienced by the footshell in the proof testing rig.

$$\sigma = \frac{F}{A} \qquad (1)$$

The assumption while using equation (1) was that the prosthetic foot would have the load applied evenly to its contact area with the floor. This is a conservative assumption, since heel strike and toe-off generate localized higher stresses that are not captured by uniform distribution. The calculations performed in A-1 resulted in a value of $412.654\,psi$ will be used to determine the eligibility of potential materials for the design in regards to their ultimate strength. Accounting for fatigue strength, a load of 344 lbf will be applied for 10×10^3 cycles, resulting in a stress of 129.44 psi. This short-term test is intended as a screening benchmark and is not equivalent to the 3-million cycle requirement.

Another parameter accounted for was the design's slip-resistance capabilities. In accordance with ACA guidelines for a Class A slip-resistance rating, the design would be subjected to a wet plane, assumed to be a linoleum tub at an incline of 12° under the same loading conditions of 344 lbs.

$$F_{friction} = N \cdot \mu \qquad (2)$$

Using equation (2), the minimum coefficient of friction was derived in A-2, equating to 0.154. However, this assumes dry conditions. For wet conditions, the coefficient of friction will be determined experimentally based on the chosen test surface, with 0.154 treated as a minimum benchmark. This value will be used as a baseline, with more conservative derivations to follow material property observations.

Solid mechanics considerations such as deflection, bending stresses, and flexural behavior are central to evaluating footshell performance. These parameters depend on defined geometric features including shell dimensions and cross-sectional profiles, which govern stress distribution under load. Fatigue analysis is equally critical because the design must withstand three million loading cycles. Since additive manufacturing materials often lack established fatigue properties, durability testing will serve as the primary method of verification. This combined theoretical and experimental approach ensures that the design's ease of installation is achieved without compromising long-term durability or safety.

Standards & Regulations:

Relevant Standards Solution Must Comply with:

1. ISO 10328: Structural testing of lower limb prostheses - Details the strength and testing requirements of lower body prosthetics. This standard includes footshell requirements and beyond sponsor requests will be the core of the project requirements. This is the only standard being required by the sponsor. https://www.iso.org/standard/70205.html [37] Maximum and minimum loads were taken from Table D.3 of 10328; 2016 using load class P6, which defines the maximum load ratings within the standard document, as requested by the sponsor.

   Static Proof Testing: Maximum load of 2490N is applied to heel and forefoot at a rate of 100-250N/s deformation null-point taken at the minimum load 50N, maximum force is held for 30±3s then deformation is compared to null-point after unloaded back to minimum load if at any point the foot fractures or deforms more than 5mm it fails the test

   Static Strength Testing: Follows similar procedure as proof testing, loading the forefoot and heel to the maximum load 4880N which is immediately unloaded after reaching maximum load. Used to find the ultimate strength of the foot at the forefoot and heel.

   Cyclic Testing: Dynamic loads are applied to the forefoot and heel up to the maximum load 1580N then unloaded to the minimum load 50N performed at a frequency between .5 and 3Hz for the number of cycles in the table. Loading is done at a rate of 100-250N/s. If any area fractures or are unable to reach the maximum load it fails the cyclic test.

Relevant Standards to Consider During Design:

2. ISO 22675: Testing of ankle-foot devices and foot units - Details further strength and testing requirements of parts that still suffered weaknesses or other complications when following the procedures of 10328. It should be noted here that testing requirements in this standard are notably less complex than in 10328, and given the sponsors wish to use 10328 abiding by this standard instead is not reasonably an option. https://www.iso.org/standard/79620.html [42]
3. ISO 10993: Biological evaluation of medical devices - is a standard that relates to quality assessment of medical devices, particularly that we shall document the product creation process thoroughly, take courses of action that reduce risk, and oversee any outsourced work where possible. In addition it focuses on the biocompatibility of medical devices in particular 10993-3 which looks at assessing long term risk of genotoxic,carcinogenic, and reproductive toxic materials, 10993-9 which assesses degradation of the device, and 10993-10 which assesses the effects on skin contact with the device. https://www.iso.org/standard/68936.html [24]
4. ISO 14971 Application of risk management to medical devices & ISO 13485: Quality management systems - Redundancy and increased specificity to the management and quality assessment aspects of 10993, where 14971 focuses on risk management and 13485 focuses on documentation and establishing who is responsible for different aspects of the projects. https://www.iso.org/standard/72704.html [43], https://www.iso.org/standard/59752.html [44]
5. CFR Title 21 Chapter I Subchapter H (Food and drugs - HHS - Medical Devices) - FDA regulations on the management of the manufacturing of medical devices. Primarily it requires a knowledgeable party to be involved in the creation in this case this would be our sponsor. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H [45]
6. EU Regulation 2017/745 - European equivalent to CFR Title 21 I-H. Important as Coyote Prosthetics sells internationally. https://eur-lex.europa.eu/eli/reg/2017/745/oj/eng [46]
7. IEC 62366: Application of usability engineering to medical devices - Centered on establishing what normal intended use is for medical devices and evaluating any risks that are associated with intended use and possible user errors. Does not relate to fixing these risks, just appropriately identifying and explaining what common risks with the device are to anyone who may use it, and mistakes that may be made to ensure it is used correctly. https://www.iso.org/standard/63179.html [47]
8. ASTM D1435-20: Standard practice for outdoor weathering of plastics - Standard relating to the weathering of plastics in the outdoors and how to appropriately test them. It is noted that weathering will vary greatly depending on location and season to season, and duration of the weathering. While it is unlikely proper weather testing will be able to be done on the solution given time constraints understanding it will be a problem is important. https://compass.astm.org/content-access?contentCode=ASTM%7CD1435-20%7Cen-US [48]

Specifications:

Spec ID	Related Need ID	Specification	Justification and Reference	Req?
A	1	Design must cause 0 injuries to prosthetic user and practitioner	Client Need	Y
B	2	Design must safely support at least 250 lbf patient weight without structural failure	Client Need	Y
C	3	Design must allow natural gait without restricting stride or altering walking motion	Client Need	Y
D	4	Design must be 2 or more pieces	Requested by Client	Y
E	5	Design must survive a repeated load of 250 lbs for at least 10,000 cycles without failing	Testing-subject's weight profile	Y
F	5	Design must survive at least 3 million cycles at 344lbf without failing	ISO 10328	Y
G	5	Design must survive a static load of 1097lbf for 30 seconds without failing	ISO 10328	Y
H	6	Design must resist slipping under 250 pounds applied to a wet 12 degree angled linoleum surface	Accountable Care Organization Guidelines	Y
I	7	Design must require no specialized tools to install	Client Need	Y
J	8,9	Under load due to 250 lbf test subject's gait, design must not shift more than 0.25 inch off of keel with/without shoe	Client Need	Y
K	10	Design must have a split between the hallux and index toe to accommodate sandals	Client Need	Y

L	11	Design must fit 28 cm foot (size US 11 Mens)	Client Need	Y
M	11	3D model of design must allow for longitudinal and lateral scalability	Client Need	Y
N	12	At least 70% of design must be created using additive manufacturing	Requested by Client	Y
O	13	Design must retain less than 30mL (2 tablespoons) of liquid after water exposure	Client Need	Y
P	14	Design must cost less than $100 per footshell to manufacture	Requested by Client	Y
Q	15	Design must make 0 alterations to prosthetic keel	Requested by Client	Y
R	16	100% of upper foot shell must be aesthetically pleasing	Requested by Client	N

Ethical and Social Analysis:

Stakeholder Identification and Ethical Analysis

The three most important stakeholders impacted by the footshell design are the prosthetic end-user who shall use the device daily, the company owner (representative of Coyote Prosthetics as a company), who is supplying and profiting from the device, and Coyote's global distributors/clients who purchase the device for sale in their own regions. An ethical analysis for each stakeholder was conducted, exploring the following ethical lenses: virtue, common good, justice, utility, rights, and care [49].

- Prosthetic Limb End-User
  - Virtue
    - Company reputation (compassion to help individuals with disabilities)
    - Retains humanity (fully functioning feet)
  - Common Good
    - Establishing normalcy for individuals with disabilities
    - Material choice (is it harmful to the environment, is it accessible to everyone)
  - Justice
    - Transparency (material usage considering allergies, fair pricing)
    - Protection from bias and exploitation against people with disabilities

- Utility
    - Improves Quality of Life
    - Limits pain and discomfort
- Rights
    - Right to safety, comfort, and human dignity

- Care
    - Designers showed compassion in the design through the emotional impact of the device for the user (comfort, aesthetics, and confidence)

- Company Owner (Coyote Prosethetics and Orthotics)
  - Virtue
    - Integrity and transparency (honoring warranty, quality products, fair pricing)
    - Motivation (improving quality of life, profit)
  - Common Good
    - Design Sustainability (material sourcing, cost, supply chain)
  - Justice
    - Fair Labor, Pricing & Marketing
    - Collaboration with clinics & medical professionals to ensure proper medical care
    - Abiding by ISO and ASME standards
  - Utility
    - Balances profit for the company with benefit for the user and the environment
    - Using analytics and impact assessments to determine the course of action when making decisions
  - Rights
    - Follows safety standards, medical device regulations, and respects intellectual property
  - Care
    - The company feels the design is an accurate representation of their company (pricing, functionality, overall help to those who need it)

- Coyote's Prosethic's International Clients/Distributors
  - Virtue
    - Charge fairly for the product across all regions
    - Act professionally with customers (Reflects on Coyote as a company)
  - Common Good
    - Equitability (regional pricing and accessibility)
    - Environmental impact (packaging, shipping methods)
  - Justice
    - Abiding by international standards and regulations (labor laws, shipping practices, and market Entry)
    - Not exploiting global markets and cheap labor
  - Utility

        - Balancing profit with global access
    - Rights
        - No counterfeiting, selling an actual product
        - Ethical handling of products and distribution of products
    - Care
        - Reliability, making sure that global users can access prosthetics

Global, Economic, Environmental, & Societal Impact

Assuming this project reaches full commercial success and enters large-scale production, its real-world implications extend far beyond the initial design and engineering process. As engineers, it becomes our professional responsibility to consider how the product affects people, industries, and the environment on a global scale. These impacts are discussed here.

Should the prosthetic foot shell design be successfully approved for global distribution, it is expected to yield a positive impact for both prosthetic users and technicians. The simplified assembly process enhances accessibility and reduces technical burden. The design's compatibility with additive manufacturing significantly lowers labor demands and minimizes the environmental impact associated with traditional shipping methods. While the sourcing and transportation of raw materials and printers remain considerations, the printer's versatility across multiple applications presents a broader benefit to global communities.

Mass manufacturing of the prosthetic foot shell presents both opportunities and responsibilities. Economically, the design's additive manufacturability reduces labor costs and increases production speed, making it feasible for clinics and distributors to offer the product at a fair price. However, ethical pricing must be maintained across regions to prevent exploitation or inequitable access to the product. The company must also ensure transparency in cost breakdowns, materials, labor, and distribution so that end-users and clinics understand what they're paying for. From a professional standpoint, honoring warranties, maintaining quality assurance, and collaborating with medical professionals ensures the product remains a trusted solution in the market. Balancing profit with accessibility is key to sustaining both company growth and user benefit.

The environmental footprint of prosthetic manufacturing is frequently overlooked, yet it poses significant challenges tied to material sourcing for additive manufacturing, energy consumption in machining, and the sustainability of labor practices. By prioritizing additive manufacturing, the design minimizes waste and reduces reliance on energy-intensive machining. Still, ethical responsibility demands understanding of material sourcing, with an emphasis on finding polymers that are recyclable, biodegradable, or derived from sustainable supply chains. Packaging and distribution methods must also be optimized to reduce carbon emissions and avoid the use of excessive plastic, especially for international clients. Long-term, the company should consider take-back programs or recycling initiatives to prevent prosthetic components from contributing to landfill waste. These efforts reflect care not only for the planet but for the communities that rely on these devices.

At its core, the prosthetic foot shell is a human-centered solution. Its success means restoring mobility, dignity, and confidence to individuals with limb loss, an impact that reaches beyond engineering. Ethical responsibilities include designing for comfort, aesthetics, and emotional resonance, ensuring users feel seen and supported through these designs. The device must be inclusive, accommodating diverse body types, skin sensitivities, and cultural preferences. Societally, the project contributes to normalizing disability and reducing stigma by offering a product that is both functional and aesthetic. Designers and manufacturers must continue engaging with users, clinicians, and advocacy groups to refine the product and uphold its role in promoting justice, care, and the common good.

Primary and Secondary Ethical Lenses

When designing any product, it is essential to consider the context of the project. In the case of a prosthetic footshell design assigned to a team of student engineers by a sponsor in the prosthetics industry, the most relevant ethical framework appears to be the common good lens. This approach prompts us to consider the full scope of connections and stakeholders affected by the design.

This lens encompasses several key relationships: the students as representatives of Boise State University, the sponsor company Coyote Prosthetics and how the final design reflects on their brand, the prosthetic users themselves, and broader public perceptions of the prosthetics industry. Given the wide-reaching impact of this project, the common good lens offers the most comprehensive and ethically sound foundation for decision-making.

Applying this lens will shape the design in multiple ways. For primary stakeholders such as Coyote Prosthetics and the end users, functionality is paramount. This forms the core of our design motivation. Additional stakeholder needs, including environmental impact and manufacturing feasibility, will be layered on top, guiding choices around materials and production methods. Ultimately, the common good lens supports a design process that maximizes net-positive outcomes across the entire stakeholder network.

In addition to the common good, we also consider the utilitarian and virtue ethics lenses. The utilitarian perspective is a natural fit for engineering projects, where practical function is always a central concern. Through this lens, we evaluate not only how the footshell performs but also how its manufacturing process affects all involved parties. For this design, utility translates into features such as tread for barefoot use, drainage to prevent waterlogging, reduced application force, and easier access to set screws. These enhancements directly benefit users, while durability benefits both users and manufacturers by minimizing the need for repairs or replacements. Additive manufacturing further supports utility by reducing cost and complexity – provided the design remains streamlined and avoids unnecessary complications. The utilitarian lens encourages us to continually ask what more we can offer, while maintaining simplicity and efficiency.

Design Description:

Functional Block Diagrams:

The level zero functional block diagram for the multi-part prosthetic foot shell, shown in Figure 7, has four inputs and eight outputs. The keel of the prosthetic keel is placed into the foot shell, providing protection against the ground or environmental forces, while also allowing the use of a shoe. The output is a user-friendly housing that transforms the keel into a fully functional foot by offering traction, enhancing stability, ensuring compatibility with footwear or barefoot use, and delivering overall user comfort.

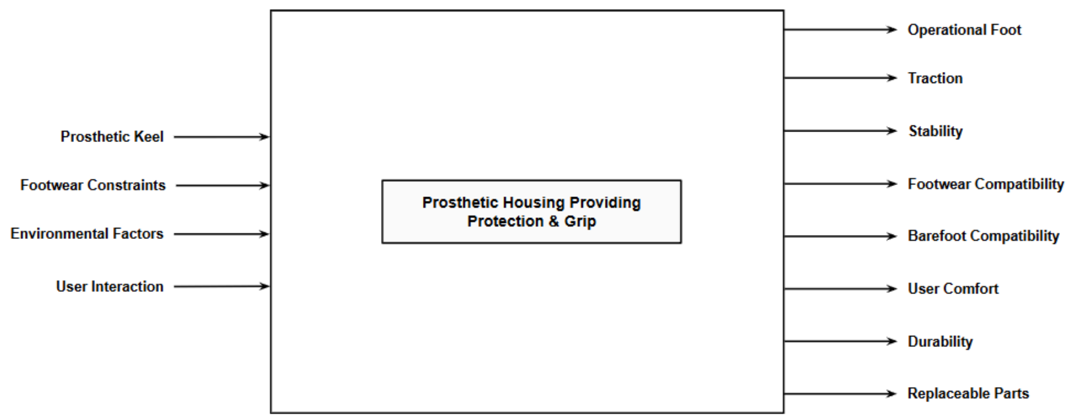


Figure 7: Level 0 Functional Block Diagram for Multi-Part Prosthetic Foot Shell

The level one functional block diagram, as shown in Figure 8 on page 30, builds upon the level zero diagram by breaking the overall function into thirteen different functions. The user interacts to attach or detach two portions of the shell assembly around the prosthetic keel, allowing the user to wear a shoe or go barefoot while still protecting the keel.

The fit of the shell assembly is determined by the fit integration function, which dictates how the footshell interfaces with the keel and external footwear. The fit integration function is broken down into a secure keel fit and common shoe size and sandal capabilities, which are dictated by the footwear constraints of the user.

The ability to easily attach and detach the front (forefoot) and rear (heel) shell parts independently while also allowing adjustment provide the user comfort. Finally, environmental inputs influence the amount of wear resistance, moisture resistance and traction mechanism the shell requires. The traction mechanism provides grip, allowing the user to go barefoot. Together, the wear resistance, moisture resistance and traction mechanism work together to output the overall durability and replaceability of components of the shell.

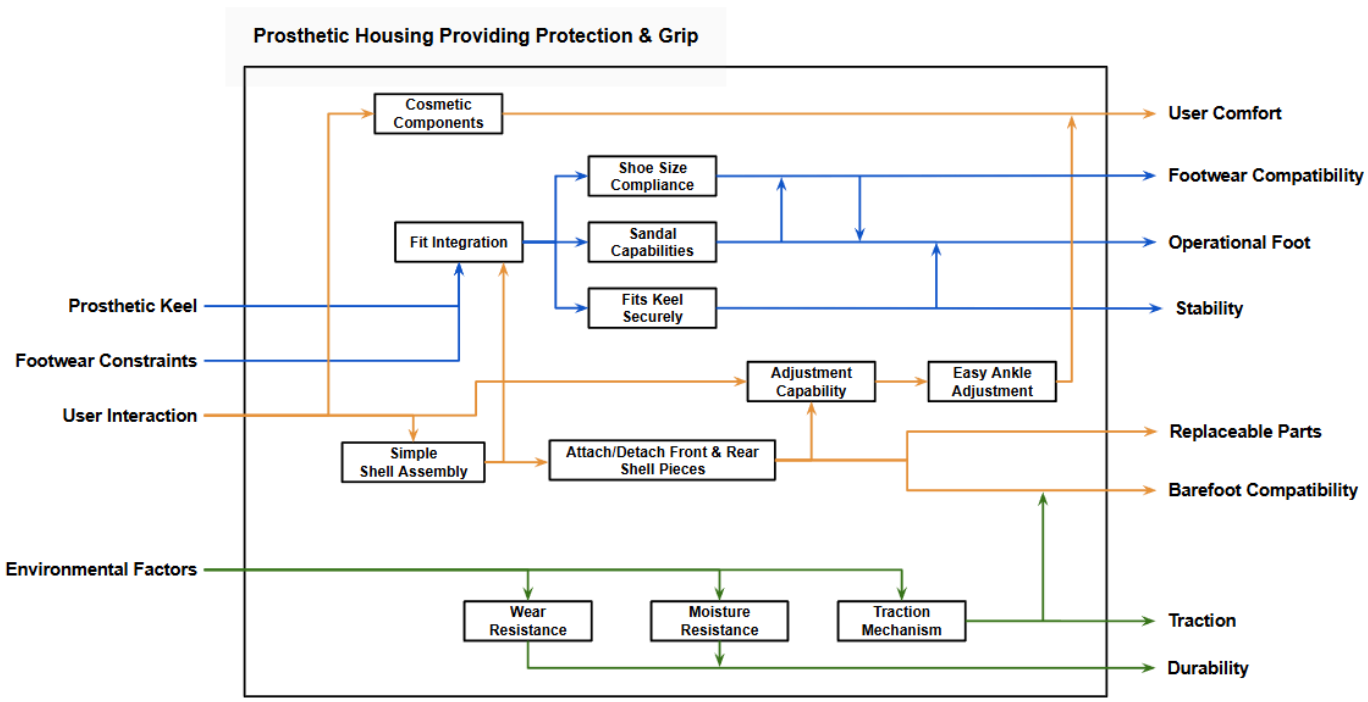


Figure 8: Level 1 Functional Block Diagram for Multi-Part Prosthetic Foot Shell

Design Description - System Level

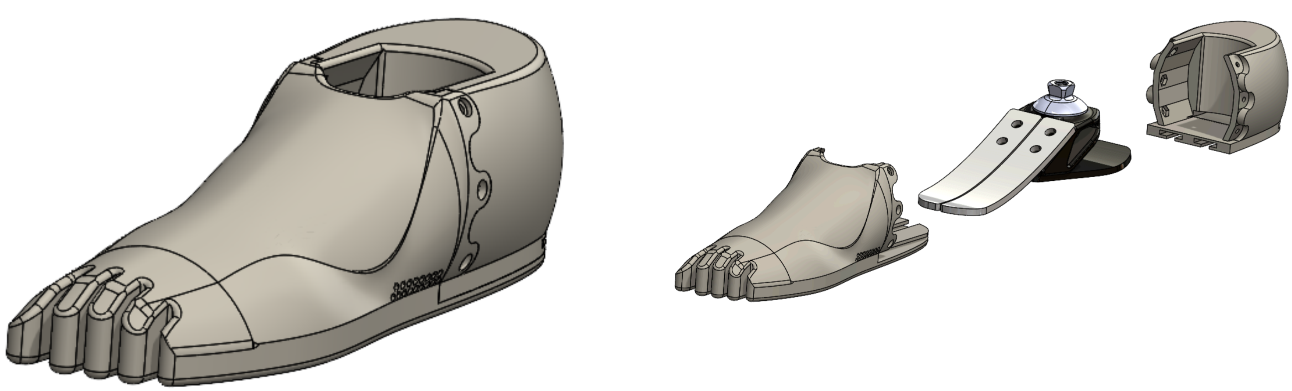


Figure 9: Footshell CAD Model (Left), Footshell Exploded View with Keel (Right)

The footshell design is divided into two main parts as shown in Figure 9, a forefoot component and a heel component. The heel piece is the main securing point that will interlock with the rear blade of the prosthetic keel (the keel of which is provided by Coyote) and becomes the anchor piece for the forefoot. The forefoot is slid onto the forward facing keel blades and attached to the heel piece. The forefoot will interlock with the heel through the use of a seam connector and rail system. A reinforced lip of molded elastomer is incorporated along the connecting seam between the forefoot and heel components, protecting the joint and redirecting shear forces from walking into the continuous shell to prevent separation. The connection between both parts of the shell will feature a geometric interlocking seam meant to increase the surface contact area between both of the parts to aid in securement. The rail system will interface between both parts to keep the heel from sliding laterally over the sole of the forefoot piece of the footshell during use. The seam will feature an interlocking design and bolt-nut connectors to lock both shell pieces together. A toe has been added to allow the use of a sandal and the bottom of the forefoot will be lightly treaded to aid grip. A cross-sectional view of the model with the prosthetic keel inserted can be seen in Figure 10 below.

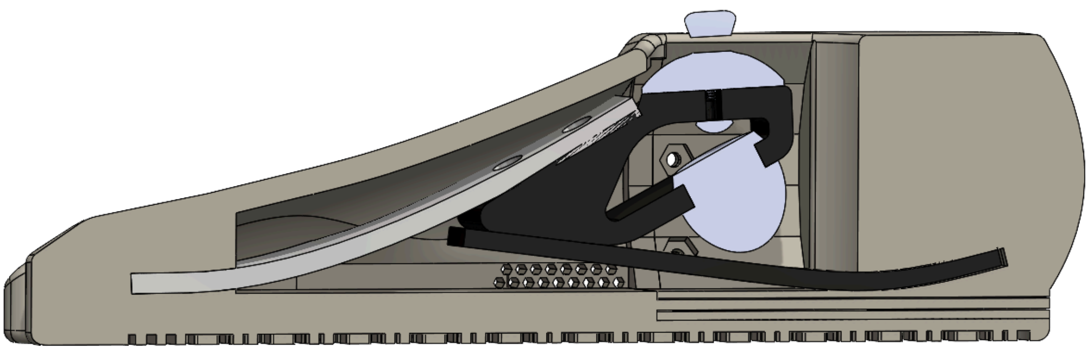


Figure 10: Cross sectional images of the footshell model showing keel interfacing

Design Description - Component Level

Forefoot

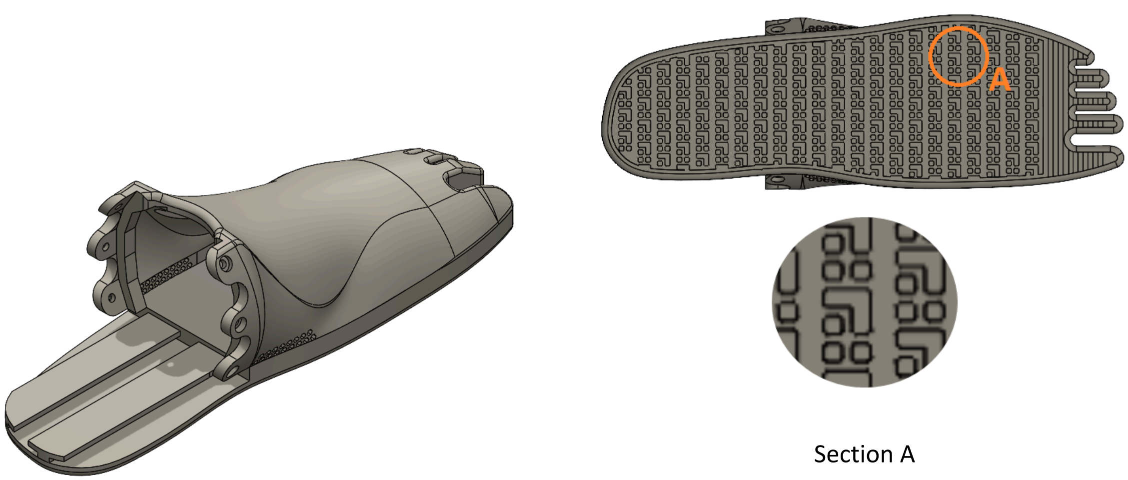


Figure 11: Isometric view of forefoot model (Left), Bottom view of Footsheel featuring tread (Right)

The forefoot features one half of an interlocking seam with a connection point being on the end of the extended sole to meet the heel piece's heel connector. In addition, there is a rail system for the bottom of the heel to slide into. Additionally it includes the front keel sleeve which is an internal pocket that will interface with the keel part of the proesthetic. A more detailed description of the rail system is included in the subsequent connections section. The forefoot will be printed with 88A shorehardness TPU (95A for initial FDM prototypes), which will allow the shell to flex as the user walks while maintaining durability compared to other 3D printing materials. The forefoot will also feature a split toe (sandal toe as described for specification I) to make the footshell compatible with flip-flops or other sandals. The sole will have a light tread as shown in Figure 11. This tread design features small inner channeling to keep liquid away from the contact surface between the shell and the ground. This increases grip without affecting the overall profile of the foot and reduces friction when trying to place the footshell in a shoe. This tread design is based on existing non-slip sole designs that are used in industrial and commercial settings. Current design meets minimum tolerances of thickness for strength of 1.5mm (0.06 inches) for printing in MJF, however, the small scale of the current tread may require tweaking as manufacturing commences and testing is performed.

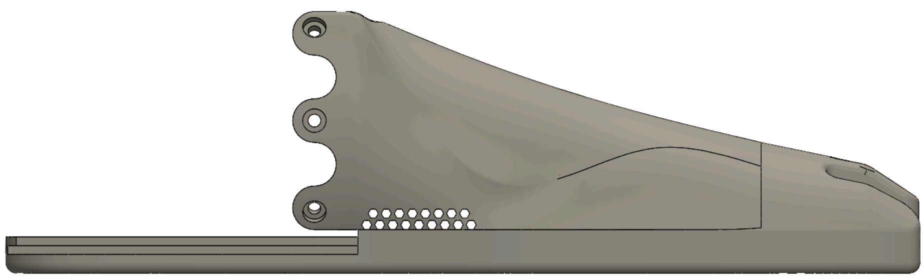


Figure 12: Left view of forefoot model showing drainage holes

To reduce waterlogging in wet environments, hexagonal drainage holes shown in Figure 12 have been added to the right and left sides in the middle section of the forefoot, which is the area of the shell where the most water will accumulate. These drainage holes allow water to escape through the footshell eliminating the need to remove the shell to empty the water. The drainage hole design was updated from the CDR from triangular shapes to reduce stress concentrations and to increase drainage.

Heel

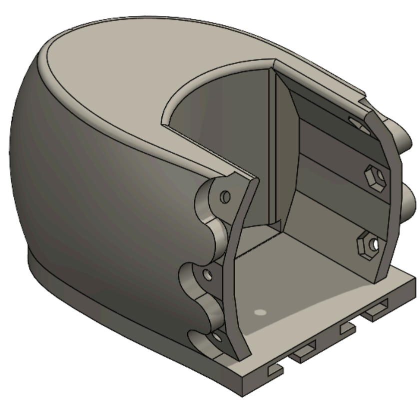
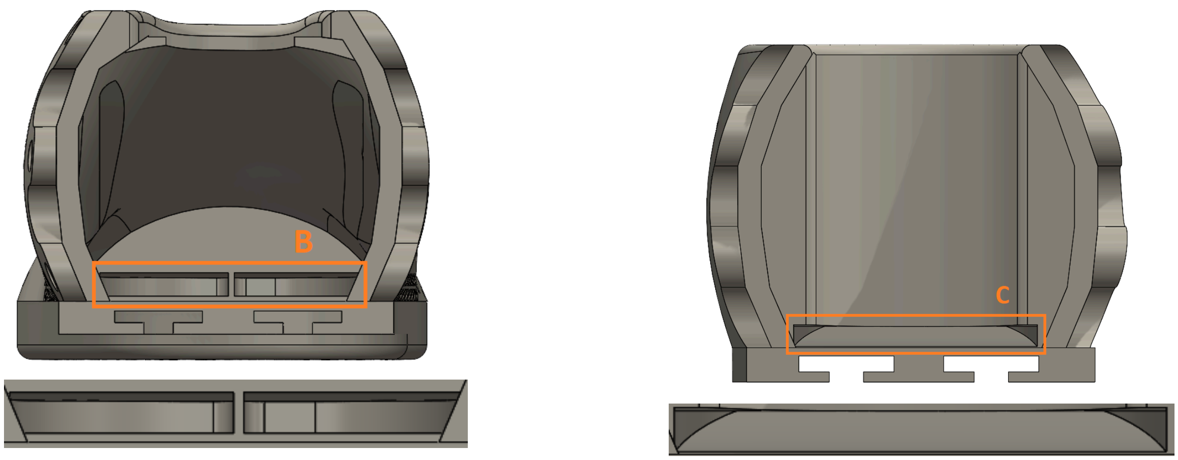


Figure 13: CAD model of heel section of the footshell

The rear, heel component of the footshell as shown in Figure 13 shall also be printed in TPU at 88A-shore level hardness for flexibility. A rear keel sleeve is embedded into the back of the component which slides over and grips the rear blade of the keel as shown in Figure 10 and Figure 14 Section C; this will be the main method of securement to the rest of the prosthetic. The heel will feature the receiving connection pieces from the forefoot, including the other half of the interlocking seam design and having T-connection grooves in the bottom to let the extended sole of the forefoot slide onto the heel via the railing system. The railing system is covered more extensively in the following interfacing section. Hexagon sockets are channeled into the seam of the heel to allow nuts to be secured into the heel of the shell. In the MJF print, a pressfit method is used for the nut securement given the high-accuracy and density capabilities. The more flexible and less precise nature of FDM printing will require a plastic binder to fully secure the nuts into the ingressed interface. A secondary securing mechanism for the rail connection is being explored and will be added to the backside of the hell should testing show that the rails alone are not fully sufficient to completely secure the heel to the sole of the forefoot without movement.

Keel Sleeves



Section B

Section C

Figure 14: Image of keel sleeve pockets on interior front shell (left) and interior heel shell (right)

The friction sleeves seen in the section views in Figure 14 take advantage of the flexibility and grip of the TPU material and will friction fit onto the keel for securement. The front sleeve is cut into the forward wall of the forefoot and features a wedge to help secure the curved, split surface at the front of the keel. The rear sleeve is slotted into a raised section in the heel portion of the shell. Both sleeve slots mirror the geometry of the keel and are easily modifiable to allow interfacing with any keel with only minor changes to dimensions.

Forefoot-Heel Interfacing

The connection points of the shell are broken into two parts, the seam and the rail system. For the seam connection, six size 6-32, 5/16" thick nuts will be pressfit into the heel piece (using epoxy as needed to secure the nuts for the FDM prints) and size 6-32, 3/8" flat head screws will be threaded through the forefoot into the nuts to form a solid connection between both parts of the shell. The bolt connections will be inset into the material on the forefoot half of the seam and feature a counter-sink to make the connection more aesthetic. Diagrams showing the bolt connection points can be seen in Figures 15 and 16 on page 34, with Figure 16 being a simplified version of the model interface (no curvature) to better show how the bolt connection works. This is an easily scalable connection that can be altered if needed by either increasing the size of the screw and nuts or increasing the number of connections around the seams. This scalability was helpful as originally the hardware design was smaller 2-56 nuts and bolts with flat heads. The smaller hardware was pulling through the material but this was easily corrected by upgrading to larder pan heads.

The rail system consists of two T-shaped components and grooves that slide and interface together between the forefoot and heel. The male end of the rail system is located on the forefoot (Figure 17i, iii on page 35) and the female end is located on the heel (Figure 17ii, iv). The forefoot T-rails will slide into heel grooves and form a secure connection to stop lateral, side-to-side movement between the sole and the heel (Figure 17v). Additional bolt and nut connections have been considered as potential fallbacks should movement still occur due to the flexibility of the TPU material, as described in the heel section. The secondary fastening mechanism, should it be required after initial testing, will use the same size 6-32 hardware as the seam connections.

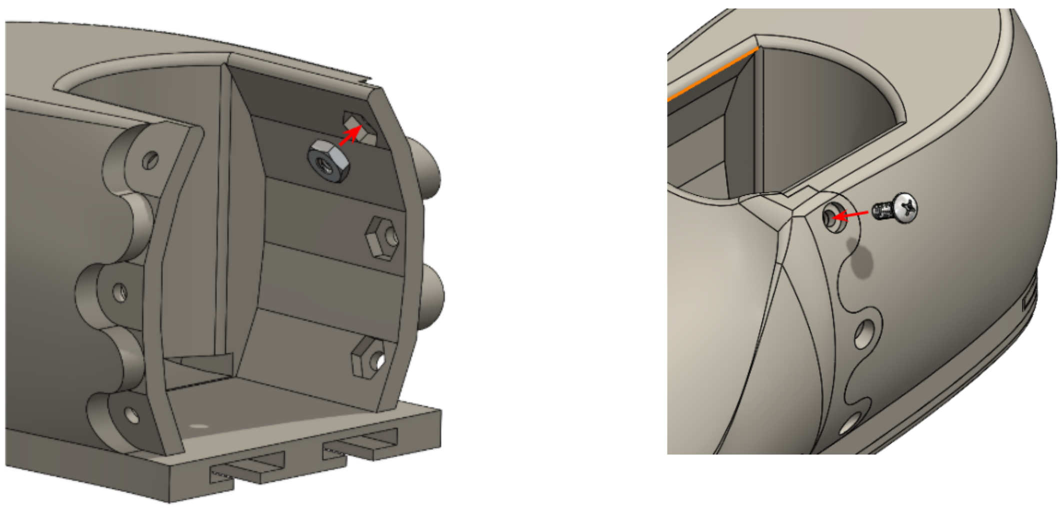


Figure 15: Interlocking Seam Design showing bolt nut and bolt connections.

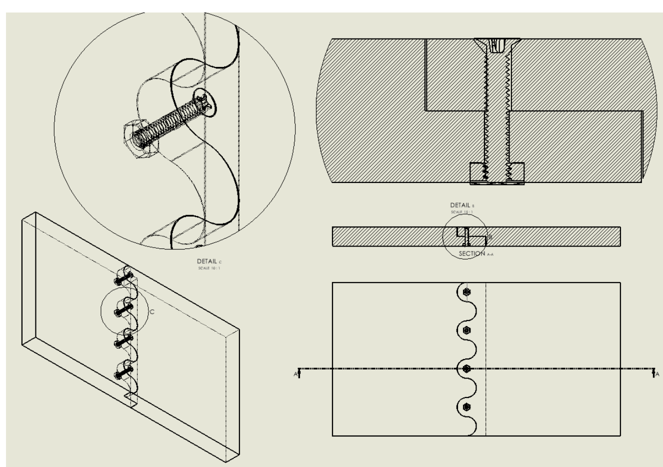


Figure 16: Wireframe drawings of simplified Interlocking seam design between forefoot and heel including cross sections for the insets for the screws and nuts.

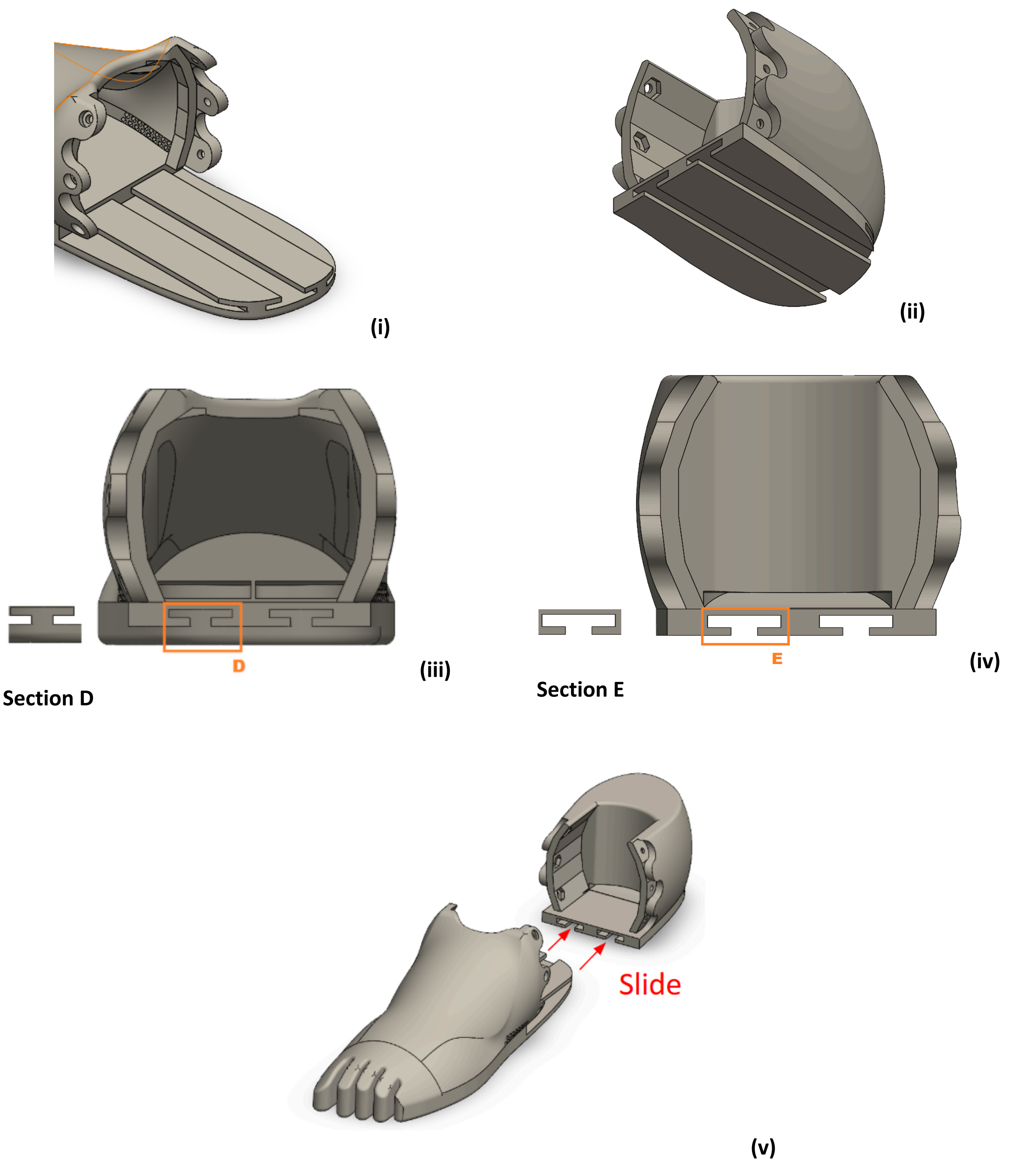


Figure 17: Isometric view of forefoot rails (i), isometric view of heel rails (ii), Section view of forefoot rail profile (iii), section view of heel rail profile (iv), image showing how forefoot and heel slide together (v).

Material and Printing Considerations

The forefoot and heel will be printed at the same time using the same material and print method. Two different print methods were selected for prototyping the design; FDM and MJF. The FDM prototypes are manufactured from 95A shore hardness TPU filament and the MJF prints from a 88A shore hardness TPU powder. For the FDM process additional holes were added (although not shown in the final design here) to allow for removal of support material that is not required for the final MJF print. The MJF process will meet the needed tolerance for the tread design and will allow intricate interfacing. It offers superior durability and strength over common FDM printing processes. This is due to MJF being a powder based with infrared light hardening. Additionally, printing with this method does not require support material throughout the print for stability which reduces wasted material and cost. Further details of the MJF process can be found in the Background and Research section on page 16. Both Boise State University and Coyote Prosthetics do not have access to MJF printers and thus a contractor will be used to complete the prints. The chosen material, 88A TPU, was the most cost effective option available from vendors that had proper balance of durability, grip, flexibility and hardness that can emulate flexibility of current molded silicon footshells. The vendor with this material, Jawstec, has been selected for prototype manufacturing by request of the sponsor given their history of business relations. Before printing in MJF through a vendor, initial prototyping will be completed through the FDM capabilities of the Coyote Prosthetics shop. This initial prototyping utilizes the most similar shore hardness level offered for FDM printing, which is 95A, and is similar to the 88A for initial manufacturability testing. This will allow early identification of potential areas of failure of the print and sizing constraints for fit to the keel and in a shoe. However, tolerances will need to be carefully examined as FDM and MJF are very different 3D printing processes. Jawstec can print TPU in MJF with an accuracy of 1.05 mm (0.04134 in) and a tolerance of ± 0.005 mm (± 0.00020 in).

Completed Prototypes

Figure 18. FDM printed footshell

Figure 19. MJF printed footshell

Printed prototypes were completed using the 95A shore hardness TPU and the MJF 88A shore hardness TPU. Images of the completed prints are shown in Figures 18 and 19 above. The FDM print worked phenomenally for testing and was liked by the client. However, the MJF prototype was unsuccessful. Further details on testing is provided in the Design Testing section on page 38. Tolerancing and final surface finish on the MJF print impeded the sliding mechanism of the rail. Filing and use of dry lube were also unsuccessful in bringing the MJF prototype to a working capacity. Additionally, the solidity of the MJF print caused the footshell to be several times heavier than expected. Successful aspects of the MJF print were the press fit for the nuts, requiring no epoxy for bonding and the lack of layer lines causing potential stress points. The lack of layer lines also gives the design a sleeker aesthetic. Overall, further iterations would be required to refine the MJF to a proper working prototype. This could be done by reducing weight through adding hollow features, or exploring lattice geometries to reduce weight while keeping structure and functionality. Future changes and recommendations will be detailed further in the Engineer's Assessment on page 45.

Design Analysis:

The specifications that are required to be mathematically guaranteed are specifications B, C, and D. Certainty is needed that the footshell will not break under the forces outlined in ISO 10328's testing parameters, so it must be able to survive a static loading of 1097lbf for 30 seconds and 3 million cycles at a cyclic loading of 344lbf. In addition, it must be verified that, under expected loading, the walls of the shell do not break due to bending stress. This will be done at 250lbf as outlined in specification B. Hand calculations are located in appendices A-3, A-4, A-5. Material properties taken from Xometry [51].

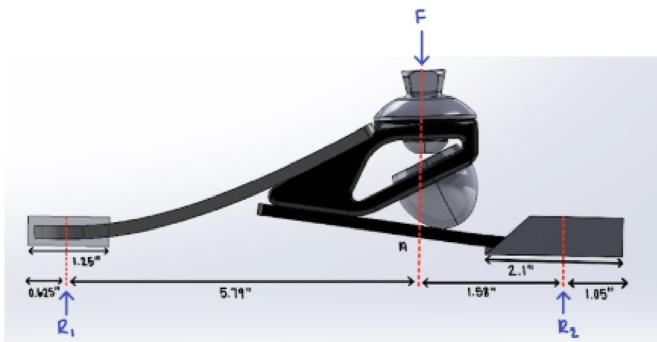


Figure 20: Free Body Diagram of Keel's Interface with Footshell Sleeves While Under Static Loading

To determine if the design can withstand the 1097 lbf load, static analysis is done on the shell. The load is applied through the ankle of the keel and two contact points seen in Figure 19, one at the interface between the toe of the keel and the footshell (R1), the other between the heel of the keel and the footshell (R2). Their contact surface areas are found in A-5. With static analysis applied, the sum of moments is equal to zero, creating equation (3).

$$\Sigma M = 0 = Fx + R1x + R2x \qquad (3)$$

Here, x is the distance from the location of the applied force to the point of study. This along with equation (4) was used to derive the expected stresses on the footshell during testing.

$$\Sigma Fy = 0 = R1 + R2 + F \qquad (4)$$

From these equations, it was determined that the heel of the footshell would experience a static loading of 860.363 psi, while the forefoot would undergo a static loading of 236.266 psi. Both loadings are well within the material tensile/compressive yielding limits of 725.69 psi.

Another necessary metric to determine the strength of the design was the structural integrity of the footshell as the proposed joints at the junction of the heel with the forefoot. These will experience the most stress under bending as force is applied. The

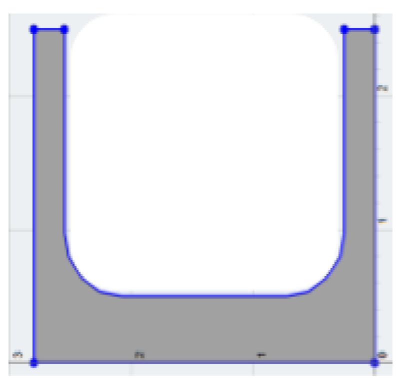


Figure 21: Simplified Cross Section of Forefoot-to-Heel Interface

cross-section was modeled as a channel shown in Figure 20, while not entirely accurate as the shell is curved, the cross section of the shell is close enough to give a good idea of the expected stresses. A simple bending moment equation (5) was employed to calculate the force applied to the junction laterally as the prosthetic-user would walk. The maximum force would be produced in the bottommost joint at the terminal stance. Observing this joint as the critical point of failure, a positive bending stress (tension) was computed (shown in A-5).

$$\sigma = \frac{M_{joint} C}{I_x} \qquad (5)$$

The point of critical failure was seen to occur at a load of 257lbs without support from the heel, creating a stress that exceeds the ultimate tensile strength of TPU. Testing for the footshell will occur with support at both the forefoot and heel, as a result during testing the moment occurring through the applied load will be much lower at an applied 257lbs than it was here. Slip-resistance was another metric verified by mathematical means. A-2 outlines the minimum necessary coefficient of friction to be 0.213 which is less than that of TPU at 0.3 [51]. Tread has been added to further ensure the prosthetic-user's safety.

Given the limited availability of reliable fatigue data for TPU, assumptions regarding fatigue strength are based on its established performance in cyclic loading applications and conservative safety margins demonstrated in static and bending analyses. While hand calculations confirm compliance with specifications B and C under static and slip-resistance conditions, fatigue strength will be verified through ISO 10328 cyclic testing to ensure long-term durability.

Design Testing:

Testing for this design is not very strenuous as most of the client needs and specifications are design requirements that are inherent in the final printed design. The specifications D, I, K, L, M, N, P, Q, and R are the specs that did not need specific testing plans and have passed through either approval or removal from our sponsor.

Test Plan C-1 (specs A, B, G) These tests are conducted by an ISO 10328-2016 testing machine. Through a series of 3 static loading tests. The first test ensures that the device can withstand 250 pounds of static load in similar conditions to a patient standing on one leg. The following two tests apply 1097 pounds of static load through each separate component (The heel and forefoot) to ensure that both parts are capable of withstanding hard impacts. Unfortunately these specs and tests are still pending as the testing machine became unavailable late into the device's testing phase. Supplemental loading scenarios were tried during client testing, showing that the device can withstand a static load of 250 pounds, but the high force scenarios are yet to be tested.

Test Plan C-2 (specs A, E, F) Similar to Test Plan C-1, this test uses an ISO 10328-2016 testing machine. Instead of simulating static loading, this is a cyclical test set for 3 million cycles. A load of 344 pounds is applied through the fully assembled keel and device part onto an angled testing surface. This surface pushes against the part at the heel and forefoot to simulate walking forces with tension and bending forces being created through the sole, and a compressive force being created at the top of the foot-shell. The test runs for three million cycles or until the device fails. Again, due to the unplanned unavailability of the testing machine, this test is still pending.

Test Plan C-3 (spec H) This test is for the slip resistant sole of the device. Under a load of 250 lbs, the device needs to not slip on a wet 6 degree linoleum surface. A ramp set at 6 degrees with a linoleum surface was created for this test and it was sprayed with soapy water, meant to simulate the conditions of a shower. During client testing of the device, they were asked to step onto the ramp and stand on the ramp with one foot. The device did not slip under this static load for 30 seconds and this spec was marked as passed. On the client's request, they asked for a tester to nudge the prosthetic while under load on the surface with their foot. Even with this added shear force the device didn't slip.

Test Plan C-4 (specs O) This test checks the functionality of the device's drain holes and water retention. Weighing the device before and a minute after full submersion in water. After submersion the device was not shook or tilted, letting the water drain through the device naturally. After running this test three times, the device held onto 22 mL of water. Being under the allowed 30mL of water, this test was marked as a success and the spec as passed.

Test Plan C-5 (specs J) This test checks the movement of the device on the keel, making sure that the footshell does not shift too much during walking. Measuring the gap between the top front lip of the footshell and the pylon after assembly, before swinging the assembly into the ground while holding the pylon 20 times. First swinging toe first measuring the shifting distance, and then swinging the device heel first and measuring again. These tests resulted in the footshell shifting by less than 1 mm, far below the allowable quarter inch of allowable shifting. This test was marked as a success and the spec was passed.

Unfortunately, specifications E,F, and G were unable to be tested due to not having access to the cycle testing apparatus required. At the start of this project it was assumed that the testing apparatus needed for these specs would be available as the project wrapped up. The situation with using the cycle tester changed too late to find another cyclic tester for use, and too late to schedule an extended client test as a replacement. These specs will be listed as pending in hopes that Coyote will test this device if they consider taking this design further. Detailed test plans are located in Appendix C.

Bill of Materials:

Table 2. Footshell Bill of Materials

Item ID	Vendor	Contract Vendor?	Part Number	Item Description	Link	Quantity	Cost/Unit ($)	Subtotal	Business Purpose
1	Jawstech	No	N/A	MJF TPU Fore-shell Piece	N/A	1	$292.05	$292.05	Additively manufactured forefoot-shell for the footshell design, including a treaded seal and geometric seam to interlock with the heel-shell piece
2	Jawstech	No	N/A	MJF TPU Heel-shell Piece	N/A	1	$234.59	$234.59	Additively manufactured heel-shell for the footshell design, including a treaded seal and geometric seam to interlock with the forefoot-shell piece and lock onto the prosthetic keel
3	McMaster Carr	No	90730A003	18-8 Stainless Steel Narrow-Profile Hex Nuts, 5/32” Width, 1/16” Height, 2-56 Thread Size, Pack of 100	https://www.mcmaster.com/90730A003	1	$4.36	$4.36	Seating connections for fastening the foreshell and heel shell pieces together.

Item ID	Vendor	Contract Vendor?	Part Number	Item Description	Link	Quantity	Cost/Unit ($)	Subtotal	Business Purpose
4	McMaster Carr	No	91771A926	Passivated 18-8 Stainless Steel Phillips Flat Head Screw, 2-56 Thread Size, 3/16"	https://www.mcmaster.com/91771A926/	1	$7.84	$7.84	Bolt connections for fastening the foreshell and heel shell pieces together.
5	McMaster Carr	No	91772A146	Passivated 18-8 Stainless Steel Pan Head Phillips Screw 6-32 Thread, 3/8" Long	https://www.mcmaster.com/91772A146/	1	$6.46	$6.46	Seating connections for fastening the foreshell and heel shell pieces together.
6	McMaster Carr	No	91841A007	Corrosion-Resistant 18-8 Stainless Steel Hex Nuts 6-32 Thread Size	https://www.mcmaster.com/91841A007/	1	$4.44	$4.44	Bolt connections for fastening the foreshell and heel shell pieces together.
7	McMaster Carr	No	7605A11	Structural Adhesive Slow-Set Epoxy, J-B Weld 8265-S, 2 FL. oz. Tube	https://www.mcmaster.com/7605A11/	1	$8.13	$8.13	Material for bonding hex nuts to the shell to prevent movement when fastening the shell-halves and during use.
8	N/A	No	N/A	TPU prints provided by Coyote Orthotics	N/A	4	$5.00	$20.00	TPU prints provided by Coyote for prototyping the shell pieces before MJF prints are ordered.
							Total Cost:	$577.87	

Bill of Materials Summary

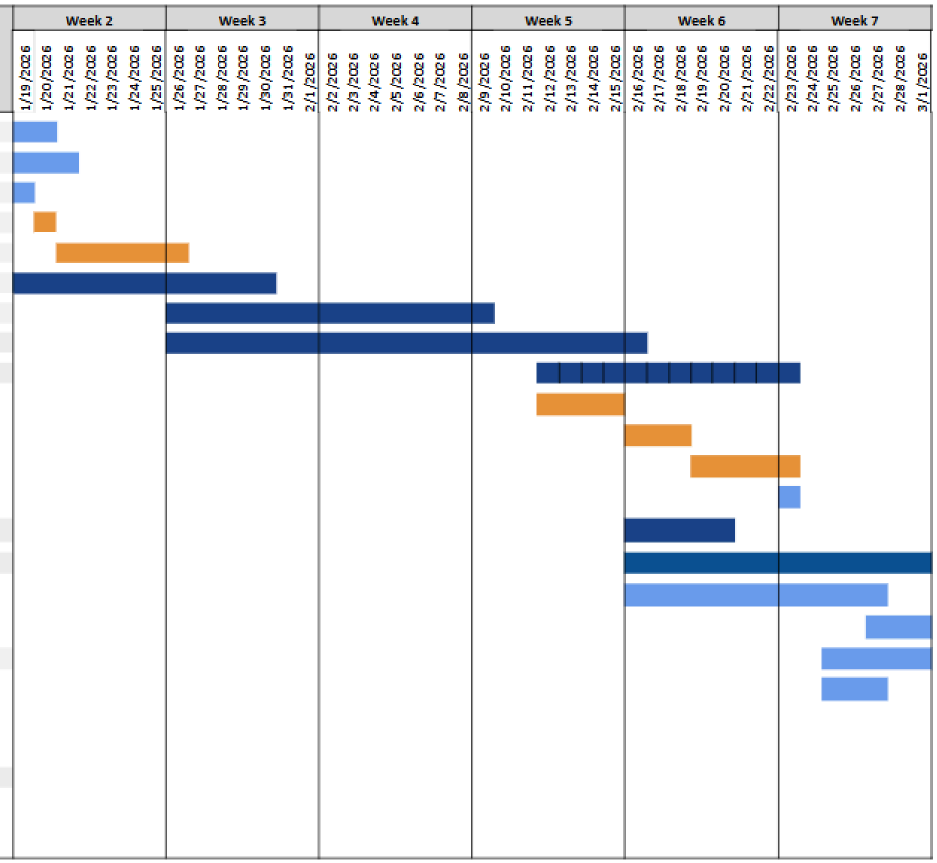
The bill of materials as shown in Table 2, is primarily derived from additive manufacturing processes with minimal cost from hardware. The CAD model determines a significant portion of the overall cost. Additive manufacturing cannot be completed through contract vendors, and the bolts and hex nuts through the contract vendors were not slim enough for the application.

Overall, costs are below the $2000 dollar budget for the project, however, the current singular footshell cost is above the $100 dollar per footshell cost requested (Specification K). However, after discussion with the sponsor, it was agreed that when ordered in bulk it would likely meet the 100 dollar or less requirement through volume discount (specification K). Due to the bulk of the project being additively manufactured, the risks involved with ordering can be broken down as follows. First, the turn around times for additive manufacturing, which are based upon Jawstech lead times, are roughly ten to thirteen business days and do not include shipping time. Thus, it can be expected that any changes necessary for our prototyping phase will require at least two weeks of lead time to produce. Second, the relative size of the footshell largely dictates the cost. The foot shell has a combined volumetric area of 26.75 in^3 between both pieces to fit a Men’s US 11 foot size, which increases the total cost from both a material and print-time standpoint. These cost and lead time risks reinforce the importance of leveraging Boise State’s on-campus FDM, and Coyote Prosthetics' printing capabilities for initial TPU prototyping. However, it’s critical to recognize that FDM-printed TPU models may differ in mechanical behavior and surface finish compared to final SLS or MJF versions, even when printing the same geometry. This underscores the need to limit post-MJF design changes and to interpret early FDM prototypes with caution regarding final performance expectations.

Schedule:

Legend: General Tasks | Model/Prints | Milestone/Submission

Task	Start	End	Assigned	% Complete
Team Charter Spring Revision	1/13/2026	1/20/2026	All	100%
Initial Gantt Chart Creation	1/19/2026	1/21/2026	All	100%
Contact Wil (Update Meeting Time)	1/13/2026	1/16/2026	BE	100%
Inital FDM Print	1/20/2026	1/20/2026	JS, JH	100%
Initial Model (IM) Updates	1/21/2026	1/26/2026	JS, JH	100%
Meet with Coyote	1/19/2026	1/30/2026	All	100%
Second FDM Print	1/26/2026	2/6/2026	JS, JH, BE. CA	100%
Finalized Test Plan & Risk Assesment	1/26/2026	2/16/2026	JW, CA, BE	100%
Further Model Updates & Iteration	2/12/2026	2/23/2026	JS, JH, All	100%
Update Sleeves	2/12/2026	2/15/2026	JS, JH	100%
Update Bolts	2/16/2026	2/18/2026	JS, JH	100%
3rd FDM Print (Print + Pickup)	2/19/2026	2/23/2026	BE	100%
Team Meeting + Design Redirections	2/23/2026	2/23/2026	All	100%
CATME (1)	2/16/2026	2/20/2026	All	100%
Showcase Abstract	2/16/2026	3/1/2026	All	100%
Rough Draft	2/16/2026	2/27/2026	CA, BE, JW	100%
Teamate QA	2/27/2026	3/1/2026	All	100%
FDM Client Testing - Delayed	2/25/2026	3/22/2026	All	100%
Contact Sponsor for Scheduling	2/25/2026	2/27/2026	All	100%
Perform Client Testing	3/18/2026	3/18/2026	All	100%
Analyze Results + Redesign	3/18/2026	3/22/2026	JS, JH	100%
Showcase Brochure	3/2/2026	3/15/2026	All	100%
Rough Draft	3/2/2026	3/8/2026	BE	100%
Teamate QA	3/9/2026	3/15/2026	All	100%



*Task titles indicated with bold/grey fill, subtasks with no fill underneath
Time lines a month from Weekly Update submission are subject to change.

Figure 22. *Gantt chart illustrating weeks 1-7 of the 2026 project timeline for the Coyote Prosthetics Footshell Capstone, including key deliverables, communication milestones, and report submissions.*

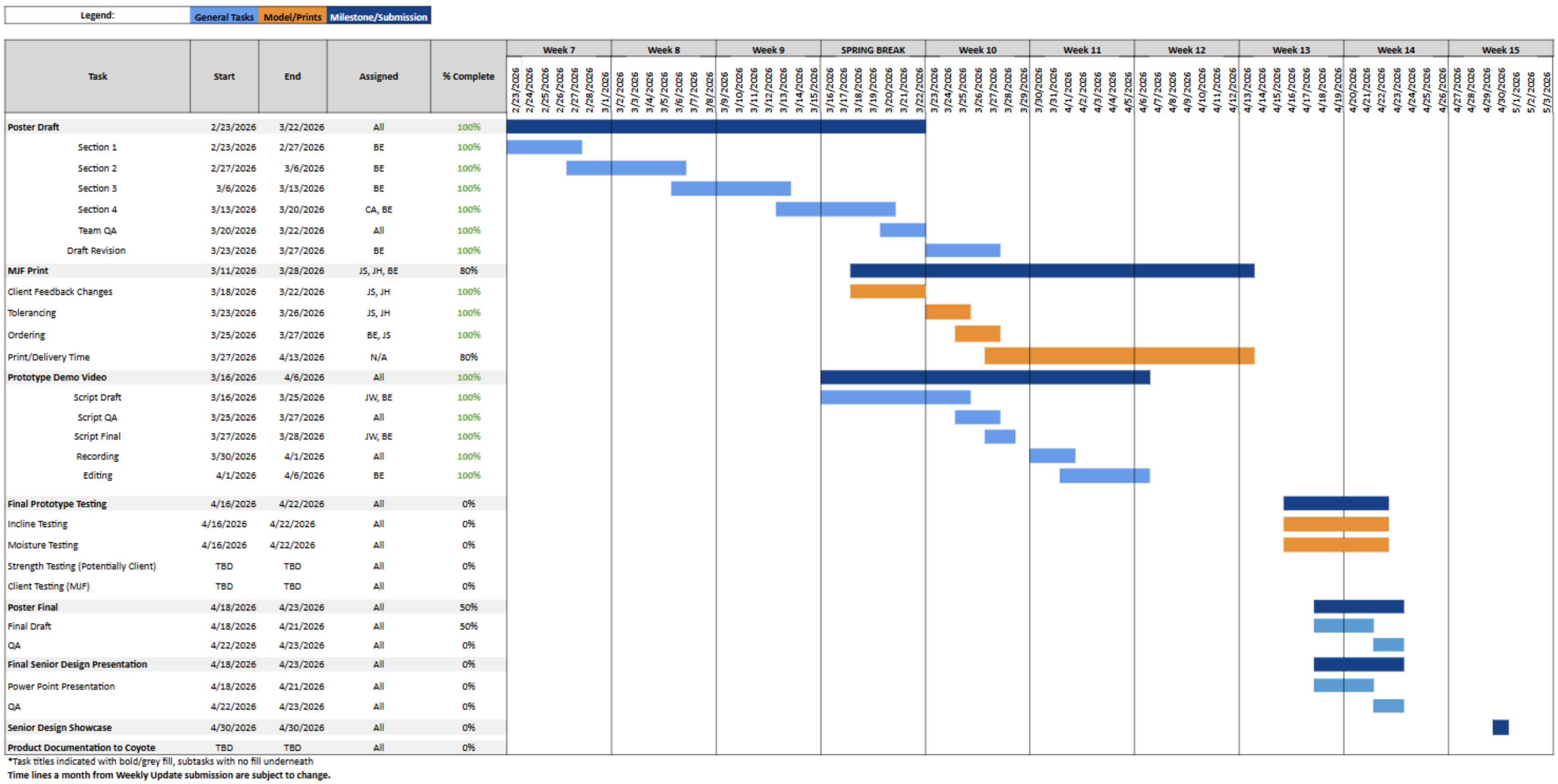


Figure 23. *Gantt chart illustrating weeks 7-15 of the 2026 project timeline for the Coyote Prosthetics Footshell Capstone, including key deliverables, communication milestones, and report submissions.*

The Coyote Prosthetics footshell design project has been streamlined to ensure a workflow that maximizes time allowed for testing and iterative processes. Approving a functioning 3D model with Coyote Prosthetics takes precedence in the manufacturing process. Once approved, material allocation and sourcing manufacturers will be necessary to proceed with testing in both a prosthetic-user and testing environment. The main point of concern remains the outsourcing of additive manufacturing as cost approval and manufacturing bottlenecks can be anticipated. Efficiency in modeling iterations and communication with manufacturers will therefore be crucial to allow enough time for testing as the final fatigue testing phase is estimated to take approximately two weeks at the maximum cycle parameters

Engineers' Assessment:

The final design successfully demonstrates the feasibility of a multi-part, additively manufactured prosthetic footshell that improves usability and expands functionality beyond current commercial models. The project's primary objective, creating a footshell that is easier to install than existing one-piece shells, was achieved through the development of a two-part TPU system secured by friction sleeves, a geometric seam, and a rail-and-bolt interface. FDM prototypes validated the core concept: the shell could be installed without tools, maintained a secure fit during gait-simulated loading, and provided adequate flexibility and comfort. These results confirm that the design direction is sound and that the modular architecture meaningfully improves clinician and user experience.

However, several shortcomings emerged during final prototyping, particularly with the MJF-printed version. The increased rigidity and surface roughness of MJF TPU caused the rail system to bind, preventing full assembly even after filing and lubrication. The MJF print was also significantly heavier than expected, indicating that internal geometry must be redesigned to reduce mass, potentially through hollowing, lattice structures, or variable-thickness walls. These manufacturability challenges highlight the need for tolerance adjustments, revised interfacing geometry, and weight-reduction strategies before the design can be considered production-ready.

From a structural standpoint, analytical calculations identified the seam and connection points as the most likely failure locations under bending loads. While TPU's elasticity likely reduces peak stresses compared to brittle-material assumptions, fatigue behavior remains uncertain due to limited published data. Full ISO 10328 cyclic testing could not be completed within the project timeline, leaving long-term durability unverified. Future iterations should incorporate fatigue-oriented geometry changes, such as thicker radii, reinforced rail shoulders, or alternative fastening strategies, to improve reliability under repeated loading.

Cost analysis revealed that single-unit MJF production exceeds the $100 target, but sponsor feedback indicates that bulk manufacturing will reduce cost to acceptable levels. This suggests that the design is economically viable at scale, though further simplification of geometry may reduce material volume and print time.

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Appendix A

A-1. Axial Stress (Keel-Ground Contact) Calculation

$$\sigma = \frac{F}{A} = \frac{1096.629\ lbf}{2.6575\ in^2} = 412.654\ psi$$

A-2. Coefficient of Friction Calculation

Solving for μ:

$$F_{friction} = N \cdot \mu = F_y cos(12) \cdot \mu = 250cos(12) \cdot \mu = 244.536\mu\ lbf$$

$$F_{slip} = F_y sin(12) = 250sin(12) = 51.978\ lbf$$

$$\mu = \frac{F_{slip}}{F_{friction}} \frac{51.978\ lbf}{244.536\ lbf} = 0.213$$

A-3. Design Analysis Calculations for Keel Sleeve Contact Area

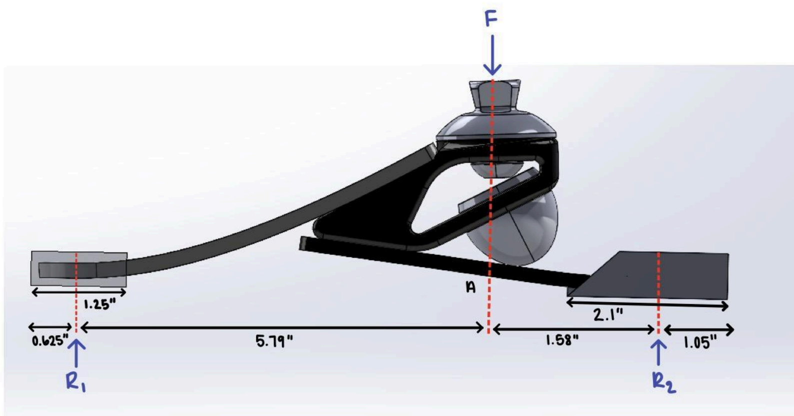
Heel Sleeve
$2.00in \cdot 2.1in = 4.2in^2$

Contact Area of Toe
$0.75in \cdot 2.21in = 1.6575in^2$

Heel Contact
$2.00in \cdot 1.29in = 2.58in^2$

Area of Toe Sleeve
$2.21in \cdot 1.25in = 2.7625in^2$

A-4. Static Analysis of Keel Sleeve Contact Stress



Static Analysis- Midstance

$\Sigma f_y = 0 \longrightarrow 1096.629\,lbf - R_1 - R_2 = 0$

$R_1 + R_2 = 1096.629\,lbf$

$\Sigma M_A = 0 \longrightarrow -R_1\left(\frac{5.79in}{12}\right) + R_2\left(\frac{1.59in}{12}\right) = 0$

$R_1\left(\frac{5.79in}{12}\right) = R_2\left(\frac{1.59in}{12}\right)$

$R_1 = R_2\left(\frac{1.59in}{5.79in}\right)$

$1096.629\,lbf = R_2\left(\frac{1.59in}{5.79in}\right) + R_2$

$R_2 = 860.363\,lbf$

$R_1 = 1096.629\,lbf - 860.363\,lbf$

$R_1 = 236.266\,lbf$

Stress

$A = 6.963\,in^2$

$\sigma = \frac{F}{A} = \frac{1096.624\,lbf}{6.963\,in^2}$

$\sigma = 157.494\,psi$

A-5. Stress Analysis of Footshell Connection Points

$$I_{x_1} = \frac{bh^3}{12} = \frac{2.8*3^3}{12}$$

$$I_{x_2} = \frac{2.3*2.5^3}{12}$$

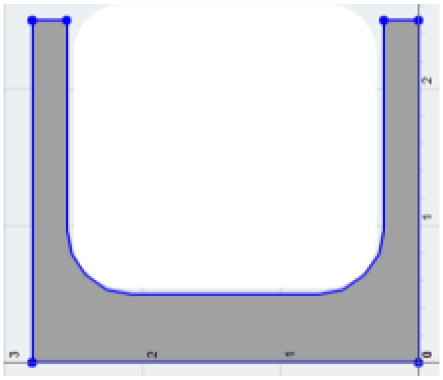
$$I_x = (I_{x_1} - (I_{x_2} + Ar^2) = 6.3 - (2.99 + (2.3 * 2.5).25^2 = 2.95in^4$$

$$c = .5y = .5 * 3 = 1.5$$

$$\sigma max = 1097psi = \frac{M*c}{I_x}$$

$$M = \frac{1097*1.5}{2.95} = 1443.4lb * in$$

$$F = \frac{1443.4}{5.595} = 258lbs$$



Appendix B

B-1. Table of alternative loading levels found in ISO 10328

Test procedure and test load			Unit	Test loading level: P8, Test loading condition: A[a]	P8 B[b]	P7 A	P7 B	P6 A	P6 B
Static test procedure	Proof test force	F_{1sp}, F_{2sp}	N	3200	3200	2900	2900	2490	2490
	Ultimate static test force	$F_{1su, lower\ level}$, $F_{2su, lower\ level}$	N	4450	4450	4100	4100	3760	3760
		$F_{1su, upper\ level}$, $F_{2su, upper\ level}$	N	5700	5700	5300	5300	4880	4880
Cyclic test procedure	Minimum test force	F_{1cmin}, F_{2cmin}	N	50					
	Cyclic range	F_{1cr}, F_{2cr}	N	2050	2050	1770	1770	1530	1530
	Maximum test force	F_{1cmax}, F_{2cmax} $F_{xcmax} = F_{xcmin} + F_{xcr}$	N	2100	2100	1820	1820	1580	1580
	Mean test force	F_{1cmean}, F_{2cmean} $F_{xcmean} = 0,5\ (F_{xcmin} + F_{xcmax})$	N	1075	1075	935	935	815	815
	Cyclic amplitude	F_{1ca}, F_{2ca} $F_{xca} = 0,5\ F_{xcr}$	N	1025	1025	885	885	765	765
	Final static force	F_{1fin}, F_{2fin} $F_{xfin} = F_{xsp}$	N	3200	3200	2900	2900	2490	2490
	Prescribed numbers of cycles		1	2×10^6					

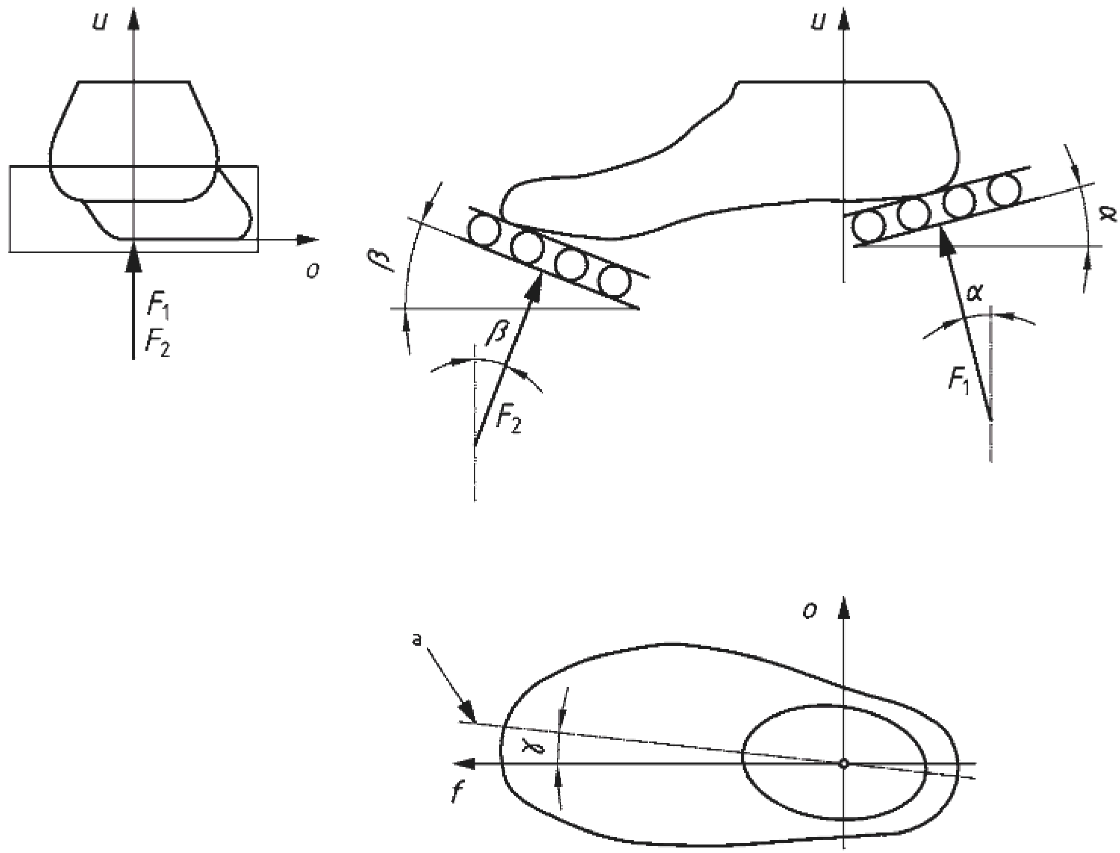
[a] Heel loading, F_{1x}.
[b] Forefoot loading, F_{2x}.

B-2. Mounting angles found in ISO 10328

Angles	Degrees
α	15
β	20
γ	7

NOTE The specified directions of loading also apply to the additional test loading levels P6, P7 and P8 specified in Annex D.

B-3. Mounting diagram found in ISO 10328



Appendix C

C-1. Testing plan for specifications A, B, G. Not Complete

Objective	Verify that the footshell can safely withstand patient-applied loads by supporting 250 lbf during normal use and enduring a 1097 lbf static load for 30 seconds, without causing injury or experiencing structural failure. ● Specification A ● Specification B ● Specification G
Assumptions	● The UTM from Coyote Prosthetics is calibrated ● The operator is trained in ISO 10328-2016 procedures ● Environmental conditions stay stable and do not affect materials ● The footshell matches final production quality and patient-use assembly ● The operator is trained in safe UTM operation and ISO-aligned methods
Hazards and PPE	Hazards ● Accidental running the testing machine during setup or adjustment. ● Sudden release of stored elastic energy in the sample during high-load testing. ● Projectile part fragments in the event of part failure ● Pinch points between the loading blocks, fixtures, and machine crosshead. PPE ● Safety glasses ● Closed-toe shoes and long pants. ● Glass/Plastic Partition
Tools required	● UTM capable of applying loads per ISO 10328-2016 ● Heel and forefoot loading blocks compliant with the standard ● Appropriate mounting fixtures for securing the footshell–keel assembly.
Test Setup, Required Environment	● Conduct testing at standard room temperature using a new, unused footshell. ● Use Coyote Prosthetic’s UTM and loading blocks. ● Position the test sample according to ISO 10328-2016 requirements, ensuring all angles, loading directions, and support platforms are aligned with the standard.

Procedure (Execution and Data Collection)	Testing procedures for these specifications fall under the testing procedures outlined in the ISO 10328 -2016 Standard [37]. As such, wording will be taken from this standard, and relevant tables and figures are added as an appendix beneath this table. 1. Prepare the part for testing by fully assembling it with a foot-ankle prosthetic keel as would be done with a user. 2. For the test in heel loading, set the angle of the direction of loading to 15 degrees, specified in Appendix B-2, and adjust the (heel) loading platform perpendicular to it. Arrange the (heel) loading platform so that it supports the forefoot, if heel loading by the test force of 560lbs deforms the test sample to such an extent that forefoot support is necessary to avoid unrealistic conditions of loading. If the test equipment uses a twin actuator set-up, ensure that the forefoot cannot contact the forefoot loading platform during heel loading 3. Mount the test sample in the test equipment, as illustrated in Appendix B-3. 4. Apply to the heel of the test sample the test force 560lbs and increase it smoothly at a rate of between 100 N/s and 250 N/s until the test sample fails, or the test force attains the value of the ultimate test force 1097lbs, without failure of the test sample. 5. Record the highest value of the test force reached during the test and whether failure has occurred. If failure has occurred end testing and do not continue unless a new part sample is available. 6. For the test in forefoot loading, set the angle of the direction of loading to 20 degrees, specified in Appendix B-2, and adjust the (forefoot) loading platform perpendicular to it. Arrange the (forefoot) loading platform so that it supports the heel, if forefoot loading by the test force of 560lbs deforms the test sample to such an extent that heel support is necessary to avoid unrealistic conditions of loading. 7. Repeat steps 4 & 5 for the forefoot rather than the heel. 8. Denote success or failure based on results

Data Analysis & Conclusion	• **PASS** if the part withstands testing on both the heel and forefoot without any structural failure. • Passing this test provides a partial pass for Spec A. Full compliance with Spec A only occurs when Specs B, E, F, G, H, I, and J are all passed. • **FAIL** if the part shows any structural failure during heel or forefoot loading. • If failure occurs, convert recorded forces to match ISO 10328 Table D.3 (P6) (Appendix A-1) if needed, and document the failure load and failure location for design revision

C-2. Testing plan for Specifications A, E, F. Not Complete

Objective	Verify that the footshell can be used safely without causing injury and can withstand a cyclic load of 344 lbf for 3 million cycles without structural failure. • Specification A • Specification E • Specification F
Assumptions	• The UTM owned by Coyote Prosthetics is assumed to be calibrated. • The test operator is trained and familiar with ISO 10328-2016 procedures. • Environmental conditions remain stable throughout testing and do not influence material behavior. • The footshell is representative of final production quality and assembled exactly as it would be for patient use. • The operator is trained in safe UTM operation and ISO-aligned testing procedures.
Hazards and PPE	Hazards • Accidental activation of the UTM during setup or alignment. • Sudden release of stored elastic energy during high-load testing. • Projectile fragments in the event of catastrophic part failure. • Pinch points between fixtures, loading blocks, and the machine crosshead. • Trip hazards from cables or equipment around the testing area. PPE • Safety glasses • Closed-toe shoes and long pants. • Glass/Plastic Partition

Tools required	- UTM capable of applying loads per ISO 10328-2016 - Heel and forefoot loading blocks compliant with the standard - Appropriate mounting fixtures for securing the footshell–keel assembly.
Test Setup, Required Environment	- Conduct testing at standard room temperature using a new, unused footshell. - Use Coyote Prosthetic's UTM and loading blocks. - Position the test sample according to ISO 10328-2016 requirements, ensuring all angles, loading directions, and support platforms are aligned with the standard.
Procedure (Execution and Data Collection)	Testing procedures for these specifications fall under the testing procedures outlined in the ISO 10328 -2016 Standard [37]. As such, wording will be taken from this standard, and relevant tables and figures are added as an appendix beneath this table. - Prepare the part for testing by fully assembling it with a foot-ankle prosthetic keel as would be done with a user. - Record the test loading level to be applied, together with the corresponding values of the angles 15 and 20 degrees at the heel and forefoot respectively and the test forces 344lbs, determining the directions and magnitudes of heel and forefoot loading, and the prescribed number of cycles. - Mount the test sample in the test equipment as illustrated in Appendix B-3 - Apply to the test sample successively the maximum test force 344lbs to the heel and the maximum test force 344lbs to the forefoot in accordance with the values for the relevant test loading level, specified in Appendix B-1. - If the test sample sustains the successive static heel and forefoot loading at 344lbs proceed - If the test sample fails to sustain the successive static heel and forefoot loading at 344lbs, record this together with the highest value of test force reached in each direction of loading and terminate the test. - Apply to the test sample alternately the pulsating test force 344lbs to the heel and the pulsating test force 344lbs to the forefoot in accordance with the requirements of 13.4.1.2 and the values for the relevant test loading level, specified in Appendix B-1, at a frequency of between 0,5 Hz and 3 Hz for a series of cycles, to allow the test sample

	and the test equipment to "settle down" and returning each load to 10lbs with each pulse.
Data Analysis & Conclusion	● **PASS** if the part withstands testing on both the heel and forefoot without any structural failure. ● Passing this test provides a partial pass for Spec A. Full compliance with Spec A only occurs when Specs B, E, F, G, H, I, and J are all passed. ● **FAIL** if the part shows any structural failure during heel or forefoot loading. ● If failure occurs, convert recorded forces to match ISO 10328 Table D.3 (P6) (Appendix A-1) if needed, and document the failure load and failure location for design revision.

C-3. Testing plan for Specification H

Objective	Design must resist slipping under 250 pounds applied to a wet 6 degree angled linoleum surface ● Specification H
Assumptions	● Design must meet Accountable Care Organization anti-slip standards ● We are assuming that this is a level of "Non-slip" that Coyote prosthetics wants to meet the need for a non-slip footshell
Hazards and PPE	Hazards ● Test subject slipping ● Water splashing into eyes ● Potential failure of test rig PPE ● Gloves ● Safety glasses/goggles ● Closed toe shoes
Tools required	● Design being tested ● Prosthetic Keel ● Linoleum Test surface ● Spray bottle ● Test rig (2x4s, nails, hammer)

Test Setup, Required Environment	● Clear testing environment with room for testing rig and space to walk around the rig
Procedure (Execution and Data Collection)	1. Assemble testing rig to hold a linoleum surface at a 6 degree incline 2. Wet linoleum surface in a fine mist of water coating the full surface through the use of a spray bottle 3. Assemble design and prosthetic keel and have the client wear it, placing full foot on the linoleum surface. a. Mark the starting position of the device, the distance of the tip of the device's toes from the top edge of the linoleum 4. Have a tester or outside person next to the client to provide balance to the client and catch them in the event of slipping. 5. Have the client begin placing more of their weight on the foot being tested until all of their body weight is on the device. a. Mark any changes in the position of the device after each weight 6. After reaching the maximum weight, wait for 30s to account for slipping a. Mark any changes in position again 7. Unload the device and remove the design to check for any damages
Data Analysis & Conclusion	● Slight shifting of device should be expected with the properties of the material used ● Sliding of the device off of the surface entirely should be accounted as a failure of the test ● **PASS** if there is only minor, temporary shifting of the footshell relative to the keel as this is acceptable due to normal material compliance and deformation under load. ● **PASS** the device may exhibit slight repositioning during loading, provided it remains fully seated on the keel and does not exceed the allowable 0.25 in displacement limit (defined in Spec J) ● **FAIL** if there is complete loss of engagement. This includes any instance where the footshell slides off the keel or loses full contact with the mounting surface.

Table 6: Data collection sheet for Testing C-3

Trial #	Data Collected
1	.25 cm
2	.1 cm
3	.33cm
4	.4cm

C-4. Testing plan for Specification O

Objective	To determine whether the footshell retains less than 30 mL of liquid after water exposure. ● Specification O
Assumptions	● Water is tested at room temperature. ● The footshell is tested in its final configuration. ● No drying, shaking, or manual water removal after exposure. ● Excess water drains naturally through the drainage holes.
Hazards and PPE	Hazards ● Slippery surfaces due to spilled water ● Electrical hazards - if tested near powered equipment PPE ● Closed-toe shoes ● Safety Goggles ● Gloves
Tools required	● Graduated cylinder ● Digital scale ● Timer ● Water source ● Catch basin ● Towels - for cleanup
Test Setup, Required Environment	● Conduct test in a controlled room-temperature environment ● Place the footshell over a catch basin. ● Position footshell in orientation for normal use. ● Confirm the scale is calibrated ● Weigh and record the dry footshell mass

Procedure (Execution and Data Collection)	1. Weigh the dry footshell. Record the mass (M_{dry}) 2. Expose the footshell to water by pouring a measured volume evenly across its surface for a consistent duration. Record volume and duration. 3. Allow the footshell to drain freely for 60 seconds without shaking, tilting, or wiping. 4. Weigh the footshell again and record the mass (M_{wet}). 5. Calculate the mass of retained water: $M_{retained} = M_{wet} - M_{dry}$ 6. Convert retained mass to volume. 7. Repeat the test three times to ensure consistency. 8. Record all data in the testing log.
Data Analysis & Conclusion	• Compute the retained water volume for each trial. • Compute the average retained volume across all trials. • Compare each trial and the average results to the specification limit of 30 mL. • **PASS** If all trials and the average retained volume are < 30 mL. • **FAIL** If any trial or the average is ≥ 30 mL.

Table 7: Data collection sheet for C-4

Trial #	Data Collected
1	~18mL
2	~22mL
3	~26mL

C-5. Testing plan for Specification J

Objective	To determine whether under expected gait loads, the footshell maintains a secure fit by not shifting more than 0.25 in relative to the keel during walking and sprinting. ● Specification J
Assumptions	● Leverage combined with testers physical strength provides a load equal to or greater than load incurred during the natural gait.
Hazards and PPE	Hazards ● Structural failure resulting in possible plastic shards acting as a projectile PPE ● Safety glasses
Tools required	● Ruler or Caliper
Test Setup, Required Environment	● Conduct testing in an open space in or outdoors that allows safe distance for all present not testing. ● Mount the footshell to the keel as per design. Attach the assembly to the patient's prosthesis. ● Confirm the footshell is fully secure before testing
Procedure (Execution and Data Collection)	1. Once the shell is fully secure, measure the distance between the pylon and edge of the footshell along both the sagittal and coronal axis. Make note of measured distance and the points from where the measurement is being made. 2. Grab keel by end of pylon at the longest lever arm and swing the heel of the footshell into the ground 10-20 times 3. Remeasure the distance between the pylon and the edge of the footshell marking any shifting beyond a quarter inch as a failure 4. Reset the keels position in the footshell for any movements beyond 0.1in then repeat steps 2 & 3 striking the forefoot on the ground instead of the heel. 5. Repeat this process until sufficient data has been collected for both shifting under load at heel and forefoot.

Data Analysis & Conclusion	Spec J: ● **FAIL** if the ankle-to-footshell distance changes by more than 0.25 in from the baseline measurement at any point ● **PASS** if the above criteria is not met

Table 8: Data collection sheet for Test C-5

Trial #	Data Collected
1	Negligible or no shifting
2	~1mm
3	~1mm
4	Negligible or no shifting